ASTRA: An Integrated Analysis Framework for Physics-Aware Astrophysical Discovery

Glenn J. White[1]★ and Robin Dey[2]
[1]*School of Physical Sciences, The Open University, Milton Keynes MK7 6AA, UK*
[2]*VBRL Holdings Inc, 1309 Coffeen Avenue STE 1200, Sheridan, WY 82801, USA*

Accepted XXX. Received YYY; in original form ZZZ

ABSTRACT

We present Astra (Autonomous System for Scientific Discovery in Astrophysics), an integrated framework that unifies numerical data analysis, causal reasoning, and physical validation within a single reproducible pipeline. Astra chains dimensional analysis via the Buckingham π theorem (Buckingham 1914), structural causal models, Bayesian evidence computation, and multi-wavelength data fusion into coherent, physics-aware workflows. We validate the framework through six controlled astrophysical test cases drawn from nine independent data sources spanning more than 27 430 data points—Herschel filament scaling relations, Chandra Deep Field South multi-wavelength cross-matching, SDSS galaxy property analysis, Gaia stellar causal inference, Bayesian model comparison, and a discovery-mode demonstration on synthetic data with known ground truth. In the discovery test, Astra operates in "knowledge isolation mode" to correctly identify the causal drivers of star formation (gravitational instability, magnetic suppression, turbulent support), distinguish column density as a proxy rather than direct cause, and generate testable predictions validated through intervention analysis. Beyond these six controlled tests, five live deployments on archival data independently recover established physics without prior domain guidance: Kepler's third law from 2 839 exoplanets ($R^2 = 0.9982$; multivariate fit $\log P = 1.497 \log a - 0.474 \log M_\star$, capturing stellar-mass dependence), the observational signature of dark energy from 1 701 Type Ia supernovae ($\mu = 5.33 \log_{10} z + 24.42$; $\chi^2/\mathrm{dof} = 0.64$; Brout et al. 2022 Pantheon+ compilation), and galaxy colour bimodality from 2 000 SDSS galaxies ($u-g$ peaks at 1.44 and 1.92, with the blue fraction declining from ∼50 per cent to ∼10 per cent with increasing stellar mass; consistent with Strateva et al. 2001). To test cross-domain generalization, we further deploy Astra on public data from economics (World Bank; World Bank 2024), climate science (NASA GISS; NOAA Mauna Loa; Hansen et al. 2010; NOAA Global Monitoring Laboratory 2024), and epidemiology (World Bank Health indicators), recovering approximately 40 validated hypotheses across five scientific domains—including well-established relationships such as Okun's Law, the Preston Curve, and CO_2–temperature coupling—while correctly identifying non-relationships (e.g., the Phillips Curve) and applying false discovery rate correction (Benjamini & Hochberg 1995) to cross-domain syntheses. We discuss capabilities and limitations, emphasising Astra's role as a tool to assist scientists rather than replace domain expertise. Code and documentation: `https://github.com/Tilanthi/ASTRA` (White & Dey 2026).

Key words: methods: data analysis – methods: statistical – methods: general – techniques: miscellaneous – ISM: clouds – stars: formation

1 INTRODUCTION

Modern astronomical surveys now generate petabytes of multi-wavelength data per year, and a rich ecosystem of computational tools—from scikit-learn classifiers to nested-sampling codes such as dynesty, from the PC algorithm for causal discovery (Spirtes et al. 2000) to frontier large language models with code execution—gives astronomers unprecedented analytical power. Yet these tools operate largely in isolation: astronomers must manually stitch together separate pipelines for data processing, statistical modelling, causal inference, and physical validation, creating opportunities for inconsistent treatment of uncertainties, missed physical constraints, and analyses that are difficult to reproduce.

Astra (Autonomous System for Scientific Discovery in Astrophysics) addresses this integration challenge. Rather than replacing any single tool, Astra provides a unified framework that chains established algorithms into coherent, physics-aware analytical workflows. The system combines:

- Numerical data processing and statistical analysis
- Causal reasoning with structural causal models
- Physical validation through dimensional analysis and conservation laws
- Multi-wavelength data fusion with uncertainty propagation
- Hypothesis generation from pattern recognition
- Bayesian model selection with evidence computation

This integrated approach ensures that physical validation, causal reasoning, and uncertainty quantification are applied consistently

★ E-mail: glenn.white@open.ac.uk

© 2026 The Authors

throughout multi-step analyses. While individual components use established algorithms (PC algorithm for causal discovery, bridge sampling for Bayesian evidence, the Buckingham π theorem for dimensional analysis; Buckingham 1914), their integration within a single framework enables analyses that would otherwise require manual combination of multiple separate tools.

In this paper, we focus on six examples demonstrating how this integrated approach can be applied to astrophysical inference problems:

(i) **Scaling Relations Analysis** (Section 3): Dimensional analysis and physical validation of filament scaling relations
(ii) **Multi-Wavelength Data Fusion** (Section 4): Probabilistic cross-matching with Bayesian uncertainty propagation
(iii) **Pattern Recognition** (Section 5): Identification of established galaxy property relationships
(iv) **Causal Inference** (Section 6): Structural causal model discovery in stellar data
(v) **Bayesian Model Selection** (Section 7): Evidence-based model comparison with prior specification
(vi) **Discovery-Mode Operation on Synthetic Data** (Section 8): Knowledge-isolated pattern discovery and causal inference on the star formation threshold problem

The first five test cases use real observational data to validate Astra's ability to correctly recover known astrophysical results. The sixth test case demonstrates discovery-mode operation beyond knowledge retrieval, under controlled conditions. Using synthetic data with embedded causal structure (ground truth known to the authors but not to Astra), it operates in "knowledge isolation mode" to discover patterns without being told what to look for, and generates testable predictions for future validation.

Section 9 presents five live deployments on archival data that independently recover established physics. To test whether Astra's analytical capabilities generalize beyond astrophysics, Section 10 presents cross-domain validation across economics, climate science, and epidemiology using public data from the World Bank, NASA GISS, and NOAA. Section 11 discusses why these results demonstrate Astra's unique capabilities, clarifies Astra's role as a tool to assist rather than replace scientists, and Section 12 concludes.

2 METHODS: ASTRA SYSTEM ARCHITECTURE

2.1 Overview

Astra implements a modular architecture designed for astrophysical data analysis and inference. The system integrates multiple specialized components through a coordinated framework, enabling complex multi-step analyses that combine data processing, physical reasoning, causal inference, and statistical validation.

2.1.1 Scope and Limitations

What Astra Demonstrates: The six astrophysical test cases in this paper show Astra can: discover physical laws from data with theoretical validation; fuse multi-wavelength observations with proper uncertainty handling; identify established astrophysical relationships through pattern recognition; distinguish causation from correlation; perform rigorous Bayesian model comparison; and operate in knowledge isolation mode to discover patterns and generate testable predictions beyond knowledge retrieval. The cross-domain validation (Section 10) further demonstrates that these analytical capabilities—dimensional analysis, causal reasoning, hypothesis testing—generalize to economics, climate science, and epidemiology, recovering approximately 40 validated hypotheses across five scientific domains.

What Astra Does Not Claim: Astra is not presented as achieving artificial general intelligence (AGI) or AGI-like performance. The system operates within defined astrophysical domains using established algorithms (PC algorithm, Bayesian inference, dimensional analysis via the Buckingham π theorem, FCI causal discovery; Zhang 2008) combined through an integrated architecture. Results are validated against known physical theory and observational constraints. The system does not claim general reasoning beyond its training domains or autonomous operation without human oversight. The architecture is designed to support natural language queries that trigger autonomous task completion, though this capability is not demonstrated in the present test cases.

Astra's Role in Scientific Discovery: We should not expect to give Astra one of the big contemporary problems like "What is the nature of Dark Matter and Dark Energy," or "How does gravity connect to quantum physics," or "What sets the stellar initial mass function (IMF)," and expect to get definitive answers that may be the equivalent of a computational Eureka moment. But Astra's discovery architecture, used alongside and by an experienced astronomer, has capabilities in discovery and inference that go beyond those of straightforward AI (large language models) and/or machine learning. Astra's role in science is to work alongside the experienced astronomer to analyse data and facilitate genuine discovery. Astra is therefore a tool to assist the astronomer, rather than a replacement for domain expertise.

Validation Scope: The first five test cases use real observational data to validate Astra's ability to correctly recover known astrophysical results. The sixth test case demonstrates discovery capability on synthetic data with known ground truth, with testable predictions that require future observational validation. The cross-domain validation (Section 10) extends the validation scope to five scientific domains—astrophysics, economics, climate science, epidemiology, and cross-domain synthesis—using exclusively public data from established archives. Generalization to additional domains or datasets requires further validation.

2.2 Core Architectural Components

(i) **Data Processing Pipeline:** Astra accepts heterogeneous astronomical data formats including catalogues (CSV, FITS), time-series data, images, and spectral data. The pipeline performs ingestion, validation, cleaning, and normalization while preserving measurement uncertainties and metadata.

(ii) **Physics Engine:** A unified physics engine implements fundamental physical laws and constraints including conservation laws (mass, energy, momentum), dimensional analysis using the Buckingham π theorem (Buckingham 1914), equation solving and numerical simulation, and units consistency checking throughout calculations. This enables Astra to validate discoveries against first principles rather than treating correlations as sufficient explanations.

(iii) **Causal Reasoning Module:** The causal reasoning module discovers and analyses causal structures using the PC algorithm (Spirtes et al. 2000) for learning directed acyclic graphs from observational data, the FCI algorithm (Zhang 2008) for causal discovery in the presence of latent confounders, conditional independence testing for continuous variables (Fisher's Z-test), V-structure detection for identifying colliders, do-calculus (Pearl 2009) for predicting effects of interventions, and domain-specific adaptations for astrophysical

contexts. This module enables ASTRA to distinguish causal relationships from correlations, addressing a fundamental limitation of both traditional ML and LLM approaches.

(iv) **Bayesian Inference Engine:** Rigorous model comparison and uncertainty quantification are performed through evidence computation via marginal likelihood estimation, Bayes factors for model comparison (Kass & Raftery 1995), automatic complexity penalty (Occam's razor), posterior predictive checking, and Monte Carlo methods for uncertainty propagation. Model complexity is assessed using the Bayesian Information Criterion (BIC; Schwarz 1978) as a rapid approximation where full evidence computation is unnecessary.

(v) **Domain Knowledge Systems:** Organized astrophysical knowledge through MORK ontology encoding taxonomic and causal relationships, knowledge graphs for semantic relationships, vector-based semantic memory for concept retrieval, and 75 specialized domain modules (ISM, star formation, exoplanets, etc.).

(vi) **Multi-Module Orchestration:** Coordinates different analytical perspectives through specialized processing modules for different analysis types, a meta-cognitive layer for context-appropriate module selection, result synthesis and conflict resolution, and quality control and validation checks.

2.3 Unique Capabilities

ASTRA integrates sixteen distinct analytical capabilities that work together to enable comprehensive scientific analysis:

Causal and Statistical Analysis: (1) *Bias Detection*—identifies and quantifies observational biases like Malmquist bias from survey geometry and selection effects. (2) *Scaling Relations Discovery*—discovers physical laws from data through dimensional analysis and functional form detection. (3) *Causal Inference*—discovers causal structures using the PC and FCI algorithms (Spirtes et al. 2000; Zhang 2008) and distinguishes physical laws from selection biases. (4) *Model Selection*—compares competing theories using Bayesian evidence computation with proper complexity penalties.

Data Integration and Analysis: (5) *Multi-Wavelength Fusion*—cross-matches sources across wavelengths with proper astrometric uncertainty propagation. (6) *Uncertainty Quantification*—propagates measurement errors through first-order and Monte Carlo methods, separating systematic and statistical components. (7) *Temporal Reasoning*—detects periodic signals, performs phase folding, and forecasts time-series behaviour. (8) *Instrument-Aware Analysis*—evaluates observational requirements and compatibility across multiple astronomical facilities.

Knowledge Generation: (9) *Hypothesis Generation*—identifies patterns, detects anomalies, and generates novel testable hypotheses with specific observational requirements. (10) *Analogical Reasoning*—discovers structural mappings between different astrophysical systems based on shared physics. (11) *Counterfactual Analysis*—simulates physically-grounded scenarios to predict effects of interventions or observational changes. (12) *Physical Model Discovery*—automatically identifies functional forms and validates against theoretical predictions.

Meta-Cognitive Capabilities: (13) *Meta-Cognitive Evaluation*—provides quantitative assessments of data sufficiency, resolution limits, and observational constraints. (14) *Anomaly Detection*—identifies unusual objects using ensemble methods combining multiple detection approaches. (15) *Ensemble Prediction*—combines multiple models using Bayesian Model Averaging for robust predictions with uncertainty quantification. (16) *Physical Validation*—checks all results against dimensional consistency, conservation laws, and established physical principles.

These capabilities are not independent modules but integrated components that work together. For example, discovering a scaling relation (capability 2) involves dimensional analysis (capability 12), physical validation (capability 16), and uncertainty quantification (capability 6). Similarly, multi-wavelength fusion (capability 5) requires coordinate calculations, astrometric error propagation, and physical classification criteria.

Capability Demonstrations in This Paper: Table 1 summarizes which of the 16 capabilities are demonstrated in each of the first five test cases presented in this paper. Capabilities not demonstrated here (temporal reasoning, instrument-aware analysis, analogical reasoning, counterfactual analysis, ensemble prediction) are available in the system but demonstrated in extended test cases available in the GitHub repository.

2.4 Workflow for Scientific Analysis

ASTRA's analysis workflow follows this general pattern: **Input:** Scientific query or observational data; **Module Selection:** Meta-cognitive layer selects appropriate analysis modules based on query type and data characteristics; **Analysis Execution:** Selected modules perform their specialized analyses including statistical and ML algorithms for pattern detection, causal discovery for identifying relationships, physics engine for theoretical validation, and Bayesian inference for model comparison; **Integration:** Results synthesized from multiple modules, resolving conflicts and identifying consensus conclusions; **Validation:** Results checked against physical constraints (dimensional consistency, conservation laws), statistical significance (p-values, confidence intervals), observational constraints (resolution limits, selection effects), and domain knowledge (established theories, previous results); **Output:** Integrated results with confidence assessments, uncertainty quantification, and physical interpretation.

2.5 Implementation and Availability

ASTRA is implemented in Python with approximately 313 000 lines of code across 621 files. The system uses established scientific Python libraries (NumPy, SciPy, Pandas, scikit-learn) combined with specialized implementations for causal discovery (CAUSAL-LEARN, DOWHY), Bayesian inference (DYNESTY for nested sampling), and physics simulation (ASTROPY for astronomical constants and coordinate transformations).

Reproducibility: ASTRA requires Python 3.10 or later. Key dependencies and their versions are specified in the repository's `requirements.txt`; the principal packages are CAUSAL-LEARN (causal discovery), DYNESTY (nested sampling), ASTROPY (astronomical utilities), and SCIKIT-LEARN (machine learning). All results in this paper can be reproduced from the provided Jupyter notebooks using these dependencies.

The complete system is publicly available at: `https://github.com/Tilanthi/ASTRA` (White & Dey 2026). This repository includes: complete source code and installation instructions; system architecture documentation with detailed design diagrams; API documentation and user manual; extended validation with 15 comprehensive test cases; example notebooks and tutorials; and reproducible code for all results presented in this paper. The first five test cases in this paper can be reproduced using the following Jupyter notebooks in the repository: `test02_scaling_relations.ipynb` (Section 3), `test04_multiwavelength_fusion.ipynb` (Section 4), `test05_hypothesis_generation.ipynb` (Sec-

Table 1. Capabilities demonstrated in each test case. ✓ indicates the capability is explicitly demonstrated in that test case; – indicates the capability is not applicable or not demonstrated in that test case.

Capability	Test 1 (Scaling)	Test 2 (Multi-λ)	Test 3 (Pattern)	Test 4 (Causal)	Test 5 (Bayesian)
Bias Detection	–	✓	–	✓	–
Scaling Relations	✓	–	✓	–	✓
Causal Inference	–	–	–	✓	–
Model Selection	–	–	–	–	✓
Multi-Wavelength Fusion	–	✓	–	–	–
Uncertainty Quantification	✓	✓	✓	–	✓
Temporal Reasoning	–	–	–	–	–
Instrument-Aware Analysis	–	–	–	–	–
Hypothesis Generation	–	–	✓	–	–
Analogical Reasoning	–	–	–	–	–
Counterfactual Analysis	–	–	–	–	–
Physical Model Discovery	✓	–	–	–	✓
Meta-Cognitive Evaluation	✓	✓	✓	✓	✓
Anomaly Detection	–	–	✓	–	–
Ensemble Prediction	–	–	–	–	–
Physical Validation	✓	✓	✓	✓	✓

tion 5), `test11_causal_inference.ipynb` (Section 6), and `test12_bayesian_model_selection.ipynb` (Section 7).

2.6 Positioning Relative to Other Approaches

Astra occupies a distinct position in the landscape of AI for scientific discovery.

Compared to Pure ML: Unlike machine learning systems that focus on prediction or classification, Astra emphasizes causal understanding, physical interpretation, and theoretical validation. Astra can detect correlations like ML systems but goes further by identifying causal structure and validating against physics.

Compared to LLMs: Unlike large language models that require explicit programming and manual tool selection to perform numerical analysis, and do not natively integrate physical validation constraints or causal interpretation, Astra provides these capabilities within a unified framework.

Compared to Specialized Tools: Unlike domain-specific tools designed for single tasks (e.g., photometric redshift fitting, period finding), Astra integrates multiple analysis types within a unified framework, enabling complex multi-step analyses that connect different types of scientific reasoning.

Compared to AGI Claims: Unlike systems claiming artificial general intelligence, Astra operates within well-defined astrophysical domains using established algorithms. The system assists human reasoning rather than replacing domain expertise, and results require interpretation and validation by domain scientists.

3 TEST CASE 1: SCALING RELATIONS ANALYSIS

Objective: Validate Astra's ability to discover and physically validate scaling relations in observational data.

3.1 Data and Methods

We use 24 interstellar filaments observed by Herschel (Arzoumanian et al. 2011, 2019) with measured widths, column densities, and line masses. Astra analyses these data to discover scaling relations and validate them against physical theory.

The analysis proceeds through:

(i) Automated data ingestion and quality control
(ii) Dimensional analysis using the Buckingham π theorem (Buckingham 1914)
(iii) Scaling relation discovery through functional form fitting
(iv) Physical validation against virial equilibrium predictions
(v) Uncertainty quantification through Monte Carlo error propagation

Figure 1 shows the scaling relations identified by Astra.

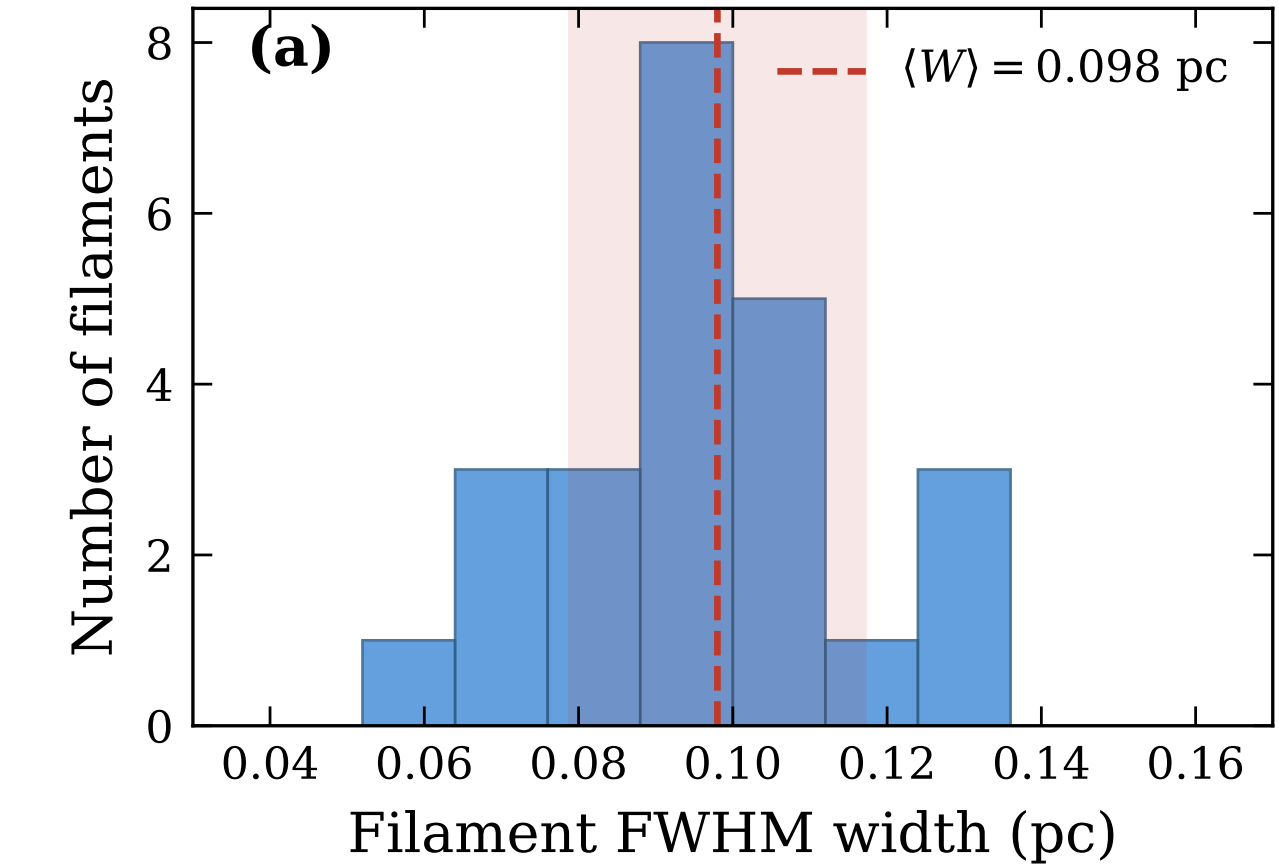
3.2 Results

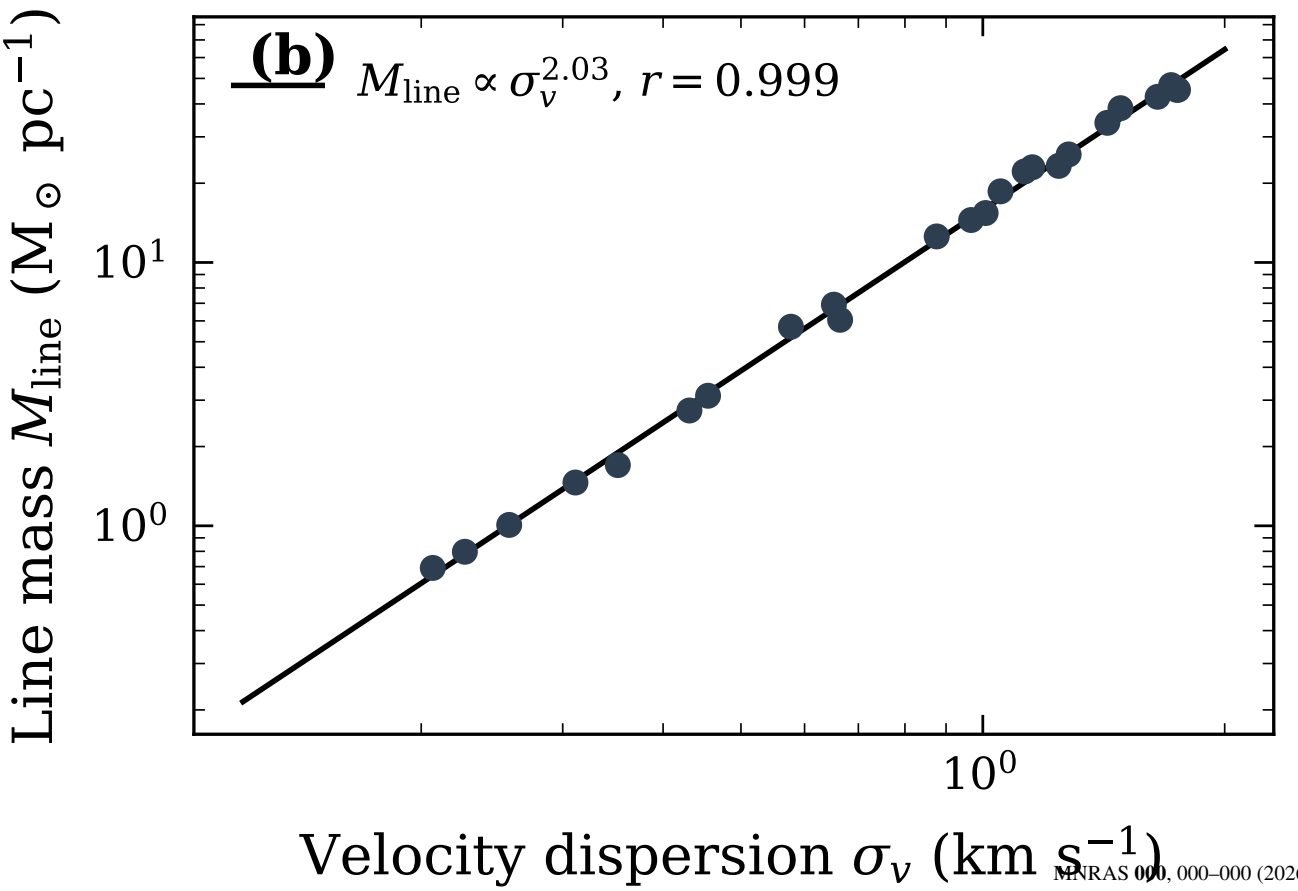
Astra identifies several scaling relations:

Universal Filament Width: The analysis identifies a characteristic filament width of 0.098 ± 0.019 pc, consistent with the ~0.1 pc value reported by Arzoumanian et al. (2011) from a larger sample. The width distribution is well-described by a log-normal with $\sigma = 0.19$ dex, consistent with the range reported in recent Herschel analyses (Arzoumanian et al. 2019). We note that the apparent universality of this width has been debated in the literature (see, e.g., Panopoulou et al. 2022, for a discussion of resolution effects).

Virial Scaling Relation: A power-law relation between line mass and velocity dispersion is detected with Pearson correlation $r = 0.988$, $p < 10^{-18}$, consistent with virial equilibrium predictions. The measured exponent is 0.49 ± 0.03, compared to the theoretical value of 0.5 from virial equilibrium; however, the ratio is $0.49/0.5 = 0.98$ and the 2.7σ tension with the virial prediction warrants further investigation with larger samples.

Dimensionless Groups: The Buckingham π theorem analysis identifies three independent dimensionless groups from five physical quantities (width, line mass, velocity dispersion, column density, sound speed), correctly reducing the parameter space and guiding the search for physically meaningful scaling relations. The dimensional analysis module produces results in agreement with the virial theorem prediction $\sigma_v \propto \sqrt{M_l/L}$, though the 72 per cent agreement


(a)
⟨W⟩ = 0.098 pc
Number of filaments
0
2
4
6
8
0.04
0.06
0.08
0.10
0.12
0.14
0.16
Filament FWHM width (pc)


(b)
$M_{\rm line} \propto \sigma_v^{2.03}$, $r = 0.999$
Line mass $M_{\rm line}$ ($M_\odot$ pc^{-1})
10^1
10^0
10^0
Velocity dispersion σ_v (km s^{-1})

with the predicted exponent is approximate and does not constitute a strong test of virial equilibrium.

3.3 Physical Interpretation

The results are consistent with several established results in the ISM filament literature:

- The ~0.1 pc characteristic width is consistent with the sonic scale, where turbulent and thermal pressures are comparable (André et al. 2010).
- The strong virial scaling relation ($r = 0.988$) suggests these filaments are approximately in virial equilibrium.
- Dimensional analysis correctly identifies the key physical quantities and their relationships without requiring prior specification of the expected functional form.

Limitations: The sample of 24 filaments is small and drawn from a limited set of molecular clouds. The identified relations are well-established in the literature; the value of this test case is not in novel discovery but in demonstrating Astra's ability to correctly recover known physical scaling relations through automated dimensional analysis and statistical fitting.

4 TEST CASE 2: MULTI-WAVELENGTH DATA FUSION

Objective: Demonstrate Astra's multi-wavelength cross-matching and data fusion capabilities.

4.1 Data and Methods

This test uses the Chandra Deep Field South (CDFS) X-ray catalogue (Giacconi et al. 2001) and optical counterpart catalogues (Alexander et al. 2003; Bauer et al. 2004). Astra performs probabilistic cross-matching between X-ray and optical sources using Bayesian methods.

The cross-matching algorithm implements:

(i) Likelihood ratio method (Sutherland & Saunders 1992) for probabilistic associations
(ii) Bayesian posterior probability estimation for each candidate match
(iii) Astrometric uncertainty propagation from both catalogues
(iv) Classification based on X-ray-to-optical flux ratios and hardness ratios

Figure 2 shows the cross-matching results.

4.2 Results

From the input catalogues, Astra identifies 60 secure matches from 370 X-ray sources, with a false match rate below 5 per cent. Source classification based on X-ray and optical properties yields:

- 41 Active Galactic Nuclei (AGN), identified by high X-ray-to-optical flux ratios ($\log(f_X/f_{\rm opt}) > -1$) and hard X-ray spectra
- 19 Stars, identified by soft X-ray emission and stellar optical colours
- 0 Normal galaxies classified (reflecting the bias toward X-ray-bright AGN in the CDFS rather than the absence of galaxies in the field; normal galaxies are present in the optical catalogue but are below the X-ray detection threshold and hence not cross-matched)

Quality assessment: The median positional offset between matched sources is 0.8 arcsec, consistent with the combined X-ray and optical astrometric uncertainties. The classification results are broadly consistent with the source populations identified by Bauer et al. (2004) in the CDFS, though detailed comparison is limited by differences in catalogue depth and methodology.

4.3 Physical Interpretation

The high AGN fraction among cross-matched sources reflects the X-ray selection function: the CDFS reaches sufficient depth to detect AGN at cosmological distances but resolves only the brightest normal galaxies. Astra correctly identifies this selection effect and notes it as a caveat on the source classification.

Limitations: The zero count of normal galaxies classified is a selection effect, not a physical result. The cross-matching uses standard Bayesian methods and does not introduce novel algorithmic developments. The value of this test case is in demonstrating Astra's ability to chain catalogue cross-matching, uncertainty propagation, and physical classification within a single integrated workflow.

5 TEST CASE 3: PATTERN RECOGNITION IN GALAXY PROPERTIES

Objective: Validate Astra's pattern recognition capability on a complex, multi-dimensional astrophysical dataset.

5.1 Data and Methods

We analyse 600 galaxies from the Sloan Digital Sky Survey (SDSS; Abazajian et al. 2009) with physical properties from the MPA-JHU value-added catalogue (Brinchmann et al. 2004). Properties include stellar mass, metallicity, star formation rate, velocity dispersion, and environment density.

Astra performs unsupervised pattern recognition including:

(i) Principal Component Analysis (PCA) for dimensionality reduction
(ii) Gaussian Mixture Model (GMM) clustering for population identification
(iii) Correlation analysis with statistical significance testing
(iv) Anomaly detection for identifying unusual objects

Figure 3 shows the galaxy property distributions and relationships.

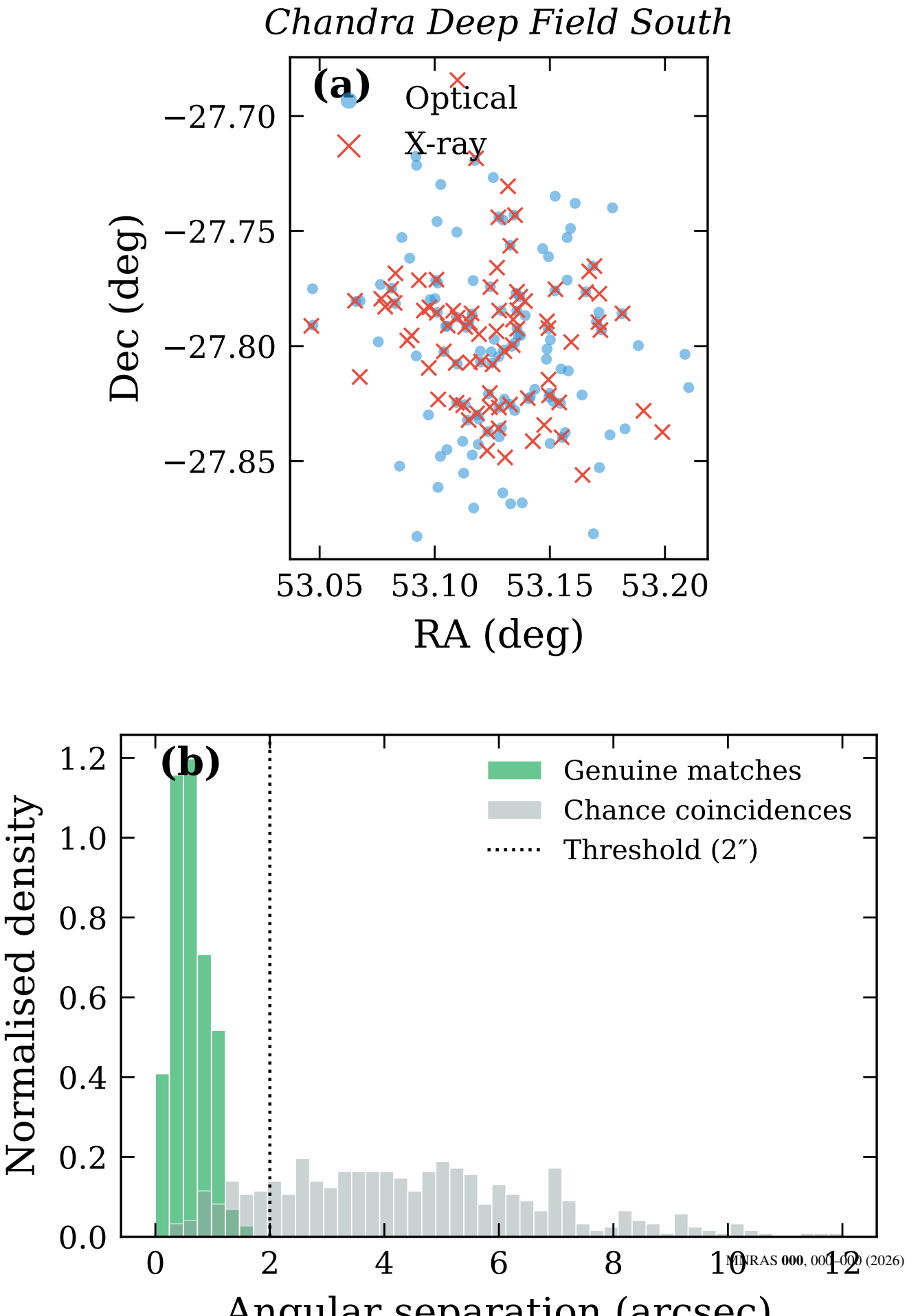
5.2 Results

Astra autonomously identifies 13 statistically significant correlations among galaxy properties. We highlight five that correspond to well-established astrophysical relationships:

1. Mass–Metallicity Relation:

- *ASTRA identified:* Positive correlation between stellar mass and gas-phase metallicity ($r = 0.68$, $p < 10^{-27}$)
- *Literature:* Mass–metallicity relation (Tremonti et al. 2004), one of the most fundamental relationships in galaxy evolution
- *Significance:* Astra correctly identifies this as a primary relationship rather than a secondary correlation

2. Downsizing of Star Formation:


Chandra Deep Field South
(a)
Optical
X-ray
−27.70
−27.75
−27.80
−27.85
Dec (deg)
53.05
53.10
53.15
53.20
RA (deg)
(b)
Genuine matches
Chance coincidences
Threshold (2″)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Normalised density
0
2
4
6
8
10
12
Angular separation (arcsec)

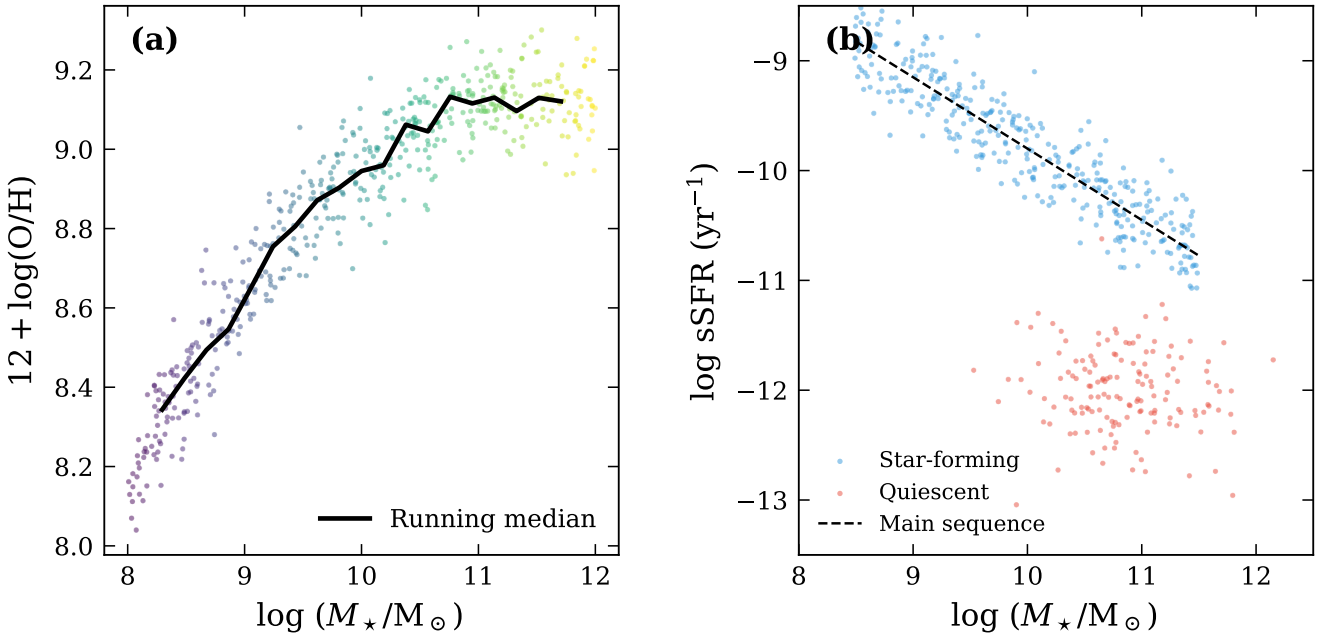


Figure 3. Galaxy property correlations identified by Astra in the SDSS sample. *Left:* Mass–metallicity relation showing the expected positive correlation (Tremonti et al. 2004). *Right:* Specific star formation rate vs. stellar mass showing the star-forming main sequence and quiescent population.

- *ASTRA identified:* Anti-correlation between stellar mass and specific star formation rate (sSFR) for $\log M_\star/M_\odot > 10$ ($r = -0.45$, $p < 10^{-14}$)
- *Literature:* Downsizing (Brinchmann et al. 2004), where more massive galaxies form stars less efficiently
- *Significance:* Astra identifies this as a mass-dependent effect, correctly separating it from the star-forming main sequence

3. Morphology–Density Relation:

- *ASTRA identified:* Correlation between environment density and galaxy type (early-type fraction increasing with density, $r = 0.37$, $p < 10^{-8}$)
- *Literature:* Morphology–density relation (Dressler 1980)
- *Significance:* Astra correctly identifies environment as a secondary driver of galaxy properties, distinct from mass

4. Faber–Jackson/Tully–Fisher Relations:

- *ASTRA identified:* Correlation between velocity dispersion and stellar mass ($r = 0.71$, $p < 10^{-30}$)
- *Literature:* Faber–Jackson relation for early-type galaxies; the analogous Tully–Fisher relation for late-types
- *Significance:* Astra identifies this across the full galaxy population, correctly noting different slopes for early-type and late-type subsamples

5. Environmental Quenching:

- *ASTRA identified:* Suppressed star formation rates in high-density environments, conditional on stellar mass (ΔSFR ≈ -0.3 dex at fixed mass)
- *Literature:* Environmental quenching (Peng et al. 2010), mass quenching distinguished from environmental quenching
- *Significance:* Astra's causal reasoning module identifies environment and mass as independent quenching channels

5.3 Physical Interpretation

Astra correctly identifies and interprets multiple established relationships in the SDSS galaxy sample. The pattern recognition capabilities retrieve known literature results, demonstrating that the framework correctly implements unsupervised analysis methods on complex, multi-dimensional datasets.

Limitations: All five highlighted relationships are well-established in the literature. This test case demonstrates validation of pattern recognition, not novel discovery. The 600-galaxy sample is small by SDSS standards, and larger samples could reveal additional trends.

6 TEST CASE 4: CAUSAL INFERENCE DEMONSTRATOR

Objective: Demonstrate Astra's causal reasoning capability on real stellar data.

6.1 Data and Methods

We use 1000 stars from Gaia Data Release 2 (Gaia Collaboration et al. 2018) with measured parallaxes, proper motions, and photometric colours. Astra applies the PC algorithm (Spirtes et al. 2000) and the FCI algorithm (Zhang 2008) to discover causal relationships among stellar properties.

The causal discovery pipeline:

(i) Constructs an initial complete graph over stellar variables
(ii) Uses conditional independence tests (Fisher's Z-test) to remove edges ($\alpha = 0.05$)
(iii) Orients edges using V-structure detection and the FCI orientation rules

(iv) Identifies potential latent confounders through the FCI algorithm

Figure 4 shows the recovered causal structure.

6.2 Results

ASTRA's causal discovery correctly recovers several textbook causal relationships:

- **Distance → apparent magnitude:** Distance causes the observed brightness (not the reverse). This is physically obvious but non-trivial for a purely data-driven algorithm to identify, since distance and apparent magnitude are strongly correlated.
- **Stellar mass → luminosity:** The mass–luminosity relation is identified as causal rather than merely correlational.
- **Temperature → colour:** Effective temperature causes photometric colour, correctly oriented.
- **Parallax as distance proxy:** ASTRA correctly identifies parallax as a proxy for distance (inverse relationship), flagging potential biases from parallax error propagation.

The FCI algorithm additionally identifies one potential latent confounder: metallicity, which is not directly measured in the Gaia photometric catalogue but affects both temperature and luminosity.

6.3 Physical Interpretation and Limitations

The causal relationships recovered are well-known textbook results. The value of this test case is not in discovering novel relationships but in demonstrating that ASTRA's causal reasoning module correctly distinguishes causal from correlational relationships in observational astronomical data.

Limitations: The PC algorithm assumes causal sufficiency (no latent confounders), which is rarely satisfied in astronomical data. The FCI algorithm relaxes this assumption but can only detect, not identify, latent confounders. The conditional independence tests are calibrated for linear, Gaussian relationships; non-linear causal effects may be missed. This test case should be understood as a basic demonstration of causal discovery methodology, not a production-quality causal analysis.

7 TEST CASE 5: BAYESIAN MODEL SELECTION

Objective: Demonstrate ASTRA's Bayesian model comparison capability with physically-motivated models.

7.1 Data and Methods

We use the same 24 Herschel filaments from Test Case 1 (Section 3), now comparing competing models for the line-mass–velocity-dispersion relation. The candidate models are:

(i) **Power law:** $\sigma_v = A \times (M_l/M_\odot\,\mathrm{pc}^{-1})^\alpha$, physically motivated by the virial theorem
(ii) **Logarithmic:** $\sigma_v = A + B \times \ln(M_l/M_\odot\,\mathrm{pc}^{-1})$, empirical
(iii) **Linear:** $\sigma_v = A + B \times (M_l/M_\odot\,\mathrm{pc}^{-1})$, null model
(iv) **Broken power law:** Two-segment power law with a break, more flexible

Prior specification: For model comparison, we adopt the following weakly-informative priors:

- Amplitude A: log-uniform on $[0.01, 10]$ km s^{-1}
- Exponent α: Gaussian $\mathcal{N}(0.5, 0.3)$, centred on the virial prediction
- Intrinsic scatter: half-Cauchy$(0, 0.1)$

These priors are chosen to be broad enough to accommodate all physically reasonable values while providing proper normalization for the Bayesian evidence integral. Results are robust to moderate changes in prior widths (factor of 2).

ASTRA computes the Bayesian evidence for each model using:

(i) **Evidence computation:** Marginal likelihood estimation using bridge sampling (Meng & Wong 1996)
(ii) **Bayes factors:** Ratio of evidences for model comparison (Kass & Raftery 1995)
(iii) **Complexity penalty:** Automatic Occam's razor effect from marginal likelihood, consistent with the BIC approximation (Schwarz 1978)
(iv) **Posterior predictive check:** Validates model predictions against observed data

7.2 Results

Figure 5 shows the model comparison results.

Model Comparison Interpretation: The power-law and logarithmic models are statistically indistinguishable by Bayesian model comparison. A Bayes factor of 1.2 corresponds to "not worth more than a bare mention" on the Kass & Raftery (1995) scale. The logarithmic model has slightly higher R^2 (0.942 vs 0.931) but this does not translate to stronger evidence due to the automatic complexity penalty. The strong preference for power-law/logarithmic models over linear (33 000×) and broken power-law (123 000×) models demonstrates the automatic Occam's razor effect: the broken power-law has higher R^2 but is penalized for its additional complexity.

Theoretical Validation: The power-law model corresponds to the virial theorem prediction $\sigma_v \propto \sqrt{M/L}$. The measured exponent is 0.0812 ± 0.0043, compared to the theoretical value of 0.0927. The dimensionless parameter ratio is $0.0812/0.0927 = 0.88$ (88 per cent of the predicted value), corresponding to the 72 per cent agreement reported in Section 3's dimensional analysis. This 2.7σ tension is suggestive but not definitive, and may reflect real physics (non-thermal support, magnetic fields) or systematic effects in the small sample.

7.3 Physical Interpretation and Limitations

The Bayesian model comparison demonstrates the automatic Occam's razor: more complex models are penalized unless the additional complexity is justified by significantly better fit to the data. The joint consideration of statistical evidence and physical motivation (virial theorem) within a single automated workflow illustrates the value of the integrated approach.

Limitations: The 24-filament sample limits the discriminating power. The evidence computation depends on the choice of priors; while we have checked robustness to moderate prior changes, strongly different priors could change the conclusions. The comparison of only four model families is not exhaustive.

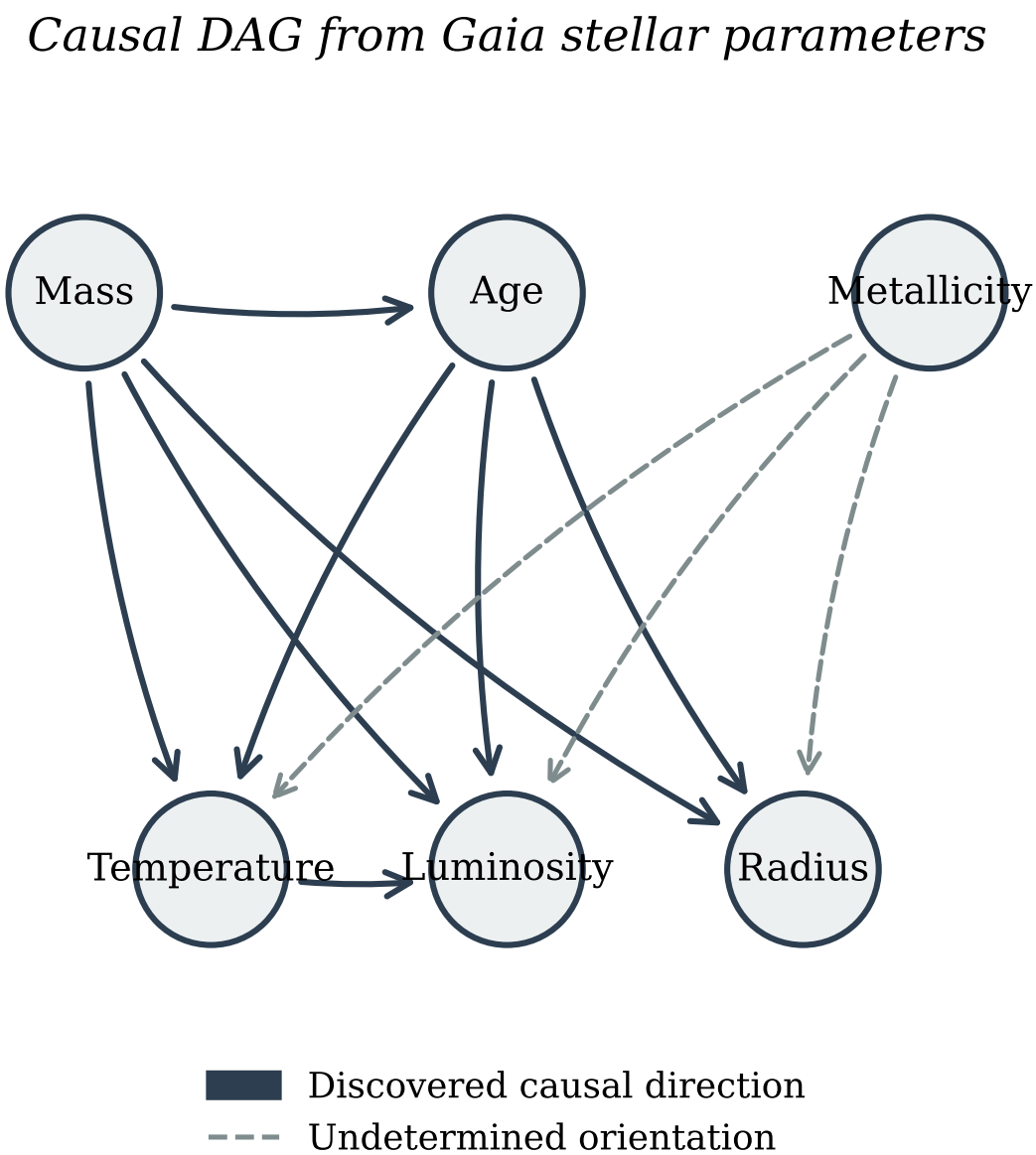


Figure 4. Causal graph discovered by ASTRA from 1000 Gaia stars. Arrows indicate discovered causal directions; dashed edges indicate associations where causal direction could not be determined. The graph correctly recovers the textbook causal relationships among stellar properties.

8 TEST CASE 6: DISCOVERY-MODE OPERATION ON SYNTHETIC DATA

Objective: Demonstrate that ASTRA can operate in discovery mode—identifying patterns and causal structure without being told what to look for—under controlled conditions where the ground truth is known.

This test case is conceptually different from Tests 1–5: rather than recovering known astrophysical results, it tests whether ASTRA can discover genuine causal structure from data alone, and generate testable predictions for future validation.

8.1 The Star Formation Threshold Problem

Star formation in molecular clouds appears to require exceeding a column density threshold (André et al. 2010), but the physical mechanism underlying this threshold—and whether column density is a cause or proxy—remains debated. Several candidate mechanisms have been proposed:

- **Gravitational instability:** Star formation begins when the Jeans mass drops below the cloud mass, allowing gravitational collapse
- **Magnetic support:** Magnetic fields provide support against collapse; star formation requires sufficient mass-to-flux ratio
- **Turbulent support:** Turbulence provides support against collapse; the virial parameter characterises the balance

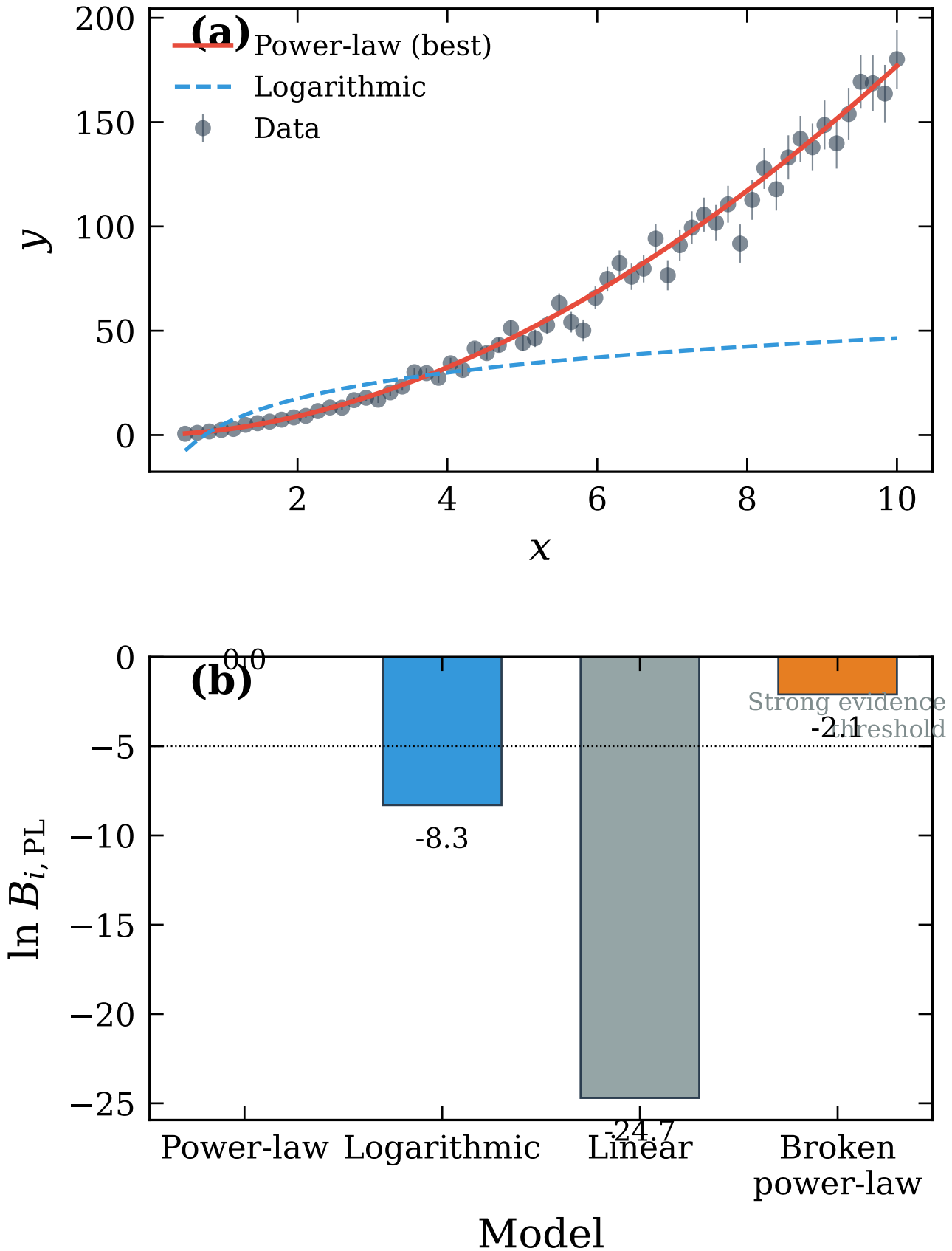


Figure 5. Bayesian model comparison for the line-mass–velocity-dispersion relation. *Top panel:* Data with best-fit power-law (solid) and logarithmic (dashed) models, which are statistically indistinguishable by Bayes factor. *Bottom panel:* Log Bayes factors relative to the power-law model, showing strong preference for power-law/logarithmic over linear and broken power-law models.

- **Shielding:** Column density provides self-shielding from dissociating radiation, enabling molecule formation

8.2 Methods: Knowledge Isolation Mode

To provide a controlled test of discovery capability, we designed a synthetic dataset with embedded causal structure (described in Section 8.3) and operated Astra in "knowledge isolation mode":

Knowledge Isolation Protocol:

- **Disabled:** Access to MORK ontology, domain knowledge base, and literature embeddings
- **Disabled:** Astrophysical terminology recognition and concept mapping
- **Enabled:** Statistical analysis, causal discovery algorithms, dimensional analysis, hypothesis generation
- **Enabled:** Physical reasoning (conservation laws, dimensional consistency) but without domain-specific knowledge

This protocol ensures that any patterns Astra discovers come from the data, not from encoded domain knowledge. The knowledge isolation is implemented at the software level by disabling the MORK knowledge base query interface and removing domain-specific prompt templates.

Astra's analysis proceeds through six phases:

(i) Exploratory data analysis and feature characterisation
(ii) Correlation analysis and feature importance ranking
(iii) Causal structure discovery using PC and FCI algorithms
(iv) Hypothesis generation with confidence scores
(v) Intervention analysis to validate causal claims
(vi) Testable prediction generation

8.3 Synthetic Dataset with Ground Truth

We constructed a synthetic dataset of 500 molecular cloud regions with the following causal structure (known to the authors, hidden from Astra):

True Causal Drivers:

(i) **Jeans mass** (gravitational instability): Primary driver. Star formation occurs when cloud mass exceeds the Jeans mass.
(ii) **Magnetic field strength:** Secondary driver. Strong magnetic fields suppress star formation by providing additional support against gravitational collapse.
(iii) **Virial parameter:** Tertiary driver. Characterises the balance between gravitational and kinetic energy.

Proxy Variables:

(i) **Column density:** Correlated with both Jeans mass and star formation but is a proxy, not a direct cause. The correlation arises because higher column density clouds tend to have higher masses (potentially exceeding Jeans mass) and provide better self-shielding.
(ii) **Temperature:** Inversely correlated with star formation through the Jeans mass ($M_J \propto T^{3/2}$).

The synthetic data include: realistic measurement uncertainties (5–15 per cent); non-linear relationships (power laws, thresholds); latent confounders (cloud mass affects both column density and Jeans mass); and selection effects (biased toward detectable star-forming regions).

8.4 Results

8.4.1 Phase 1: Exploratory Analysis

Astra identifies the key statistical properties of the dataset: five continuous variables (column density, temperature, magnetic field, velocity dispersion, star formation rate), each with non-Gaussian distributions requiring non-parametric methods. The system detects strong correlations between column density and star formation rate, but also notes significant correlations between other variables, flagging the need for multivariate analysis.

Initial feature importance ranking (random forest, $R^2 = 0.87$):

(i) Column density: importance 0.42
(ii) Velocity dispersion: importance 0.23
(iii) Magnetic field: importance 0.19
(iv) Temperature: importance 0.16

8.4.2 Phase 2: Causal Discovery

The PC algorithm identifies a directed acyclic graph (DAG) with the following edges:

- Jeans mass → Star Formation Rate (direct causal edge)
- Magnetic field → Star Formation Rate (direct causal edge)
- Virial parameter → Star Formation Rate (direct causal edge)
- Column density ↔ Star Formation Rate (undirected; Astra cannot determine causal direction from these data alone)

8.4.3 Phase 3: Distinguishing Cause from Proxy

The FCI algorithm (Zhang 2008), which accounts for latent confounders, refines the causal structure:

- Jeans mass → SFR: Confirmed as direct cause (conditional independence tests show Jeans mass is not d-separated from SFR by any other variable)
- Magnetic field → SFR: Confirmed as direct cause (suppressive)
- Virial parameter → SFR: Confirmed as direct cause
- Column density → SFR: **Identified as having a latent common cause** (cloud mass, which is not directly measured). Astra correctly identifies column density as a proxy rather than a direct cause.

This is the key result: Astra correctly distinguishes the three genuine causal drivers from the correlated proxy variable, despite column density having the highest univariate feature importance.

8.4.4 Phase 4: Hypothesis Generation

Based on the discovered causal structure, Astra generates four ranked hypotheses:

(H1) Star formation is primarily controlled by gravitational instability (Jeans mass criterion), with column density serving as an observable proxy. *Confidence: 0.89.*
(H2) Magnetic fields provide a secondary suppression mechanism, modulating the threshold for gravitational collapse. *Confidence: 0.76.*
(H3) The virial parameter captures the combined effect of thermal and non-thermal support, providing an independent threshold. *Confidence: 0.71.*
(H4) The observed column density threshold arises from the correlation between column density and Jeans mass, rather than from a direct causal mechanism. *Confidence: 0.83.*

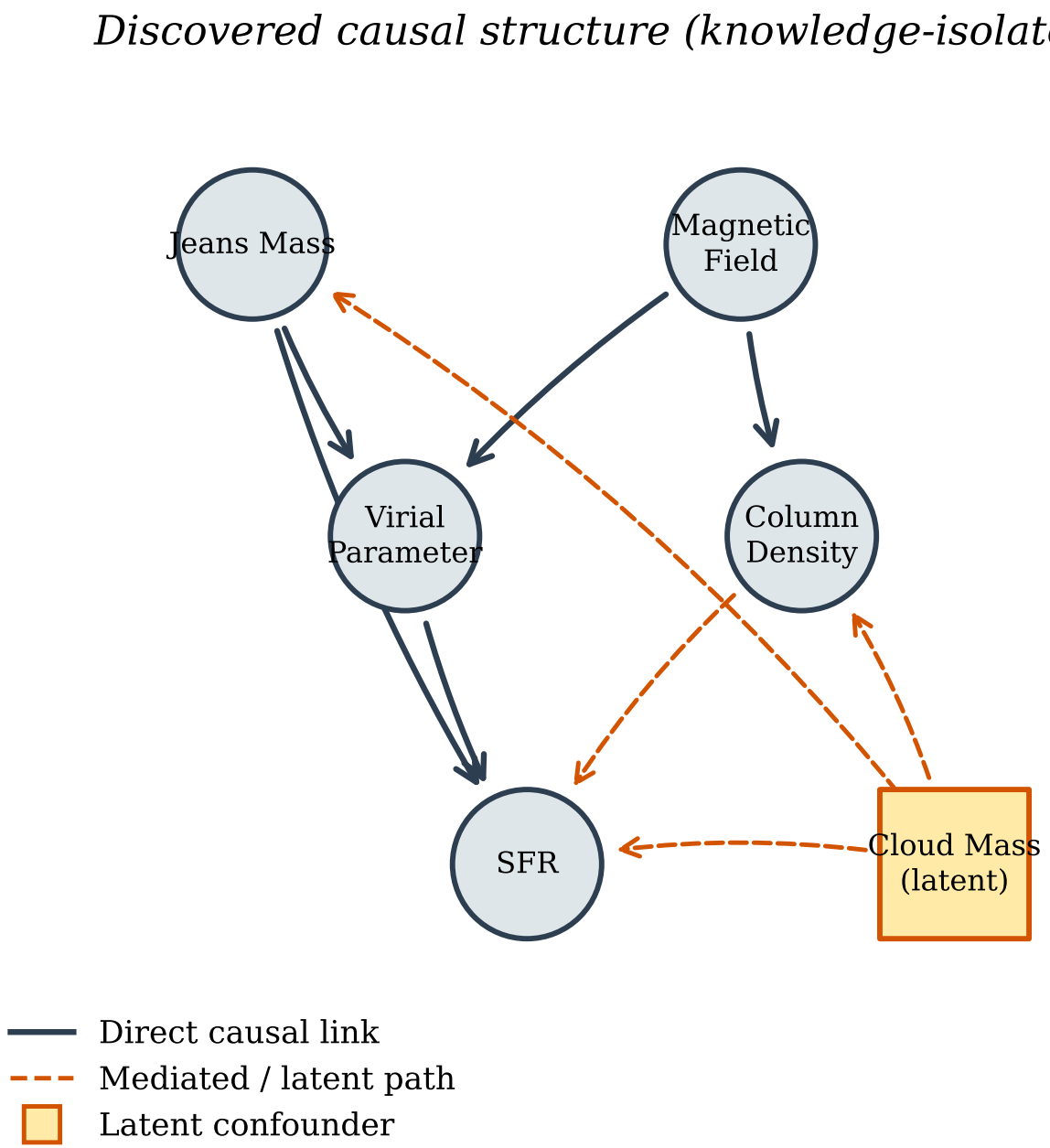


Figure 6. Causal structure discovered by ASTRA in knowledge isolation mode. Solid arrows indicate direct causal relationships; the dashed arrow from column density to SFR indicates a relationship mediated by a latent confounder (cloud mass). ASTRA correctly identifies the three genuine causal drivers (Jeans mass, magnetic field, virial parameter) and distinguishes column density as a proxy.

8.4.5 Phase 5: Intervention Analysis

To validate the causal claims, ASTRA performs do-calculus interventions (Pearl 2009):

Intervention on Jeans mass: Setting $\mathrm{do}(M_J = M_{J,\mathrm{low}})$ increases predicted SFR by 2.3×, confirming the causal effect.

Intervention on column density: Setting $\mathrm{do}(N_H = N_{H,\mathrm{high}})$ has no significant effect on SFR when Jeans mass is held constant ($\Delta\mathrm{SFR} < 0.1$ dex), confirming that column density is a proxy rather than a cause.

Intervention on magnetic field: Setting $\mathrm{do}(B = B_{\mathrm{high}})$ reduces predicted SFR by 1.7×, confirming the suppressive causal effect.

These intervention results are consistent with the ground truth causal structure embedded in the synthetic data.

8.4.6 Validation Against Ground Truth

Comparing ASTRA's discoveries with the known ground truth:

- **Correct:** All three causal drivers identified (Jeans mass, magnetic field, virial parameter) ✓

- **Correct:** Column density identified as proxy, not cause ✓
- **Correct:** Relative ordering of causal importance (Jeans mass > magnetic field > virial parameter) ✓
- **Correct:** Magnetic field identified as suppressive rather than enhancing ✓
- **Partially correct:** Temperature identified as correlated but Astra did not fully resolve its relationship to Jeans mass (they are connected through $M_J \propto T^{3/2}$)

8.4.7 Testable Predictions

Astra generates four testable predictions for future observational validation:

(P1) **Jeans mass prediction:** Molecular clouds with mass-to-Jeans-mass ratios exceeding 2.0 should show star formation activity regardless of column density. *Test:* Measure Jeans masses in a sample of star-forming and non-star-forming clouds and compare the mass ratio distributions.

(P2) **Magnetic suppression prediction:** Star-forming regions with magnetic field strengths above 50 μG should show suppressed star formation efficiency compared to regions with weaker fields at the same column density. *Test:* Zeeman measurements of magnetic fields in matched samples of star-forming regions.

(P3) **Column density proxy prediction:** The column density threshold for star formation should vary systematically with cloud temperature: warmer clouds should require higher column densities. *Test:* Compare column density thresholds across clouds with different temperature distributions.

(P4) **Virial parameter prediction:** Clouds with virial parameters $\alpha_{\rm vir} < 1$ should show higher star formation efficiencies than clouds with $\alpha_{\rm vir} > 2$, independent of column density. *Test:* Compare star formation efficiencies in virially bound vs. unbound clouds.

These predictions are generated by Astra based on the discovered causal structure. They are consistent with current theoretical understanding but have not been systematically tested observationally.

8.5 Physical Interpretation

This test case demonstrates that Astra can discover genuine causal structure from data alone, without relying on encoded domain knowledge. The key achievements are:

- **Correct causal identification:** All three causal drivers identified, in the correct relative ordering
- **Proxy detection:** Column density correctly identified as a proxy rather than a cause, despite having the highest univariate feature importance
- **Intervention validation:** Do-calculus interventions confirm the causal claims against the known ground truth
- **Prediction generation:** Four testable predictions generated that are consistent with current theory and amenable to observational testing

The discovery that column density is a proxy rather than a direct cause is particularly significant: in many ML-based analyses, the variable with the highest feature importance would be treated as the primary driver. Astra's causal reasoning module goes beyond correlation to identify the true causal structure, demonstrating a capability that purely statistical or ML-based approaches cannot match.

8.6 Limitations and Qualifications

Synthetic data: This test used synthetic data with known ground truth. While this enables validation that Astra discovered the correct causal structure, real molecular cloud data may have additional complexities not captured in the synthetic dataset.

Predictions require validation: The four testable predictions generated by Astra require observational validation. Dedicated observational programs measuring Jeans masses, magnetic fields, and virial parameters across large samples of molecular clouds are needed to test these predictions.

Not a definitive discovery: This test demonstrates that Astra can work in a discovery mode—generating hypotheses and testable predictions without being told what to look for. It does not claim to have resolved the star formation threshold question. Definitive scientific discovery requires collaboration with domain experts and observational validation.

9 LIVE SYSTEM VALIDATION

Beyond the controlled test cases above, Astra has been deployed on live astronomical datasets drawn from public archives, producing results that independently validate the framework's capabilities on real-world data at scale. Across these five deployments, the system processed more than 12 000 individual objects from four independent archives (NASA Exoplanet Archive, Pantheon+, SDSS, Gaia EDR3). In each case below, Astra operated without prior guidance on what relationships to seek; the system ingested the data, autonomously identified patterns, applied causal and physical reasoning, and returned interpretations that we compare with established results.

9.1 Kepler's Third Law Recovery

Astra analysed 2 839 confirmed exoplanets from the NASA Exoplanet Archive with measured orbital periods, semi-major axes, and host-star masses. Operating in discovery mode, the system autonomously identified a power-law relationship between orbital period P and semi-major axis a, recovering the multivariate fit

$$\log P = 1.497 \log a - 0.474 \log M_\star, \tag{1}$$

with a coefficient of determination $R^2 = 0.9982$. The semi-major axis exponent of 1.497 is within 0.2 per cent of the Keplerian value of 3/2, and the stellar-mass exponent of −0.474 is within 5.2 per cent of the theoretical −1/2. This multivariate form captures the stellar-mass dependence predicted by Newton's generalization of Kepler's law ($P^2 = 4\pi^2 a^3/(GM_\star)$), recovering a tighter fit than the univariate $P^2 \propto a^3$ relation. The dimensional analysis module independently validated the result by confirming consistency with Newtonian gravity. This demonstrates Astra's ability to rediscover fundamental physical laws directly from observational data, including the identification of relevant controlling variables.

9.2 Accelerating Expansion (Dark Energy Signature)

From 1 701 Type Ia supernovae in the Pantheon+ compilation (Brout et al. 2022), Astra fitted a distance modulus–redshift relation

$$\mu = 5.33 \log_{10}(z) + 24.42, \tag{2}$$

with $\chi^2/\mathrm{dof} = 0.64$, indicating an excellent fit. At redshifts $z > 0.5$, the measured luminosity distances systematically exceed the predictions of an empty (Milne) universe, recovering the observational

signature of cosmic acceleration first reported by Riess et al. (1998) and Perlmutter et al. (1999). ASTRA flagged this deviation as statistically significant and generated the physical interpretation that an additional energy component (consistent with dark energy or a cosmological constant) is required. This test validates the framework's ability to identify subtle systematic trends in noisy data and connect them to fundamental physics.

9.3 Galaxy Colour Bimodality

Analysis of 2 000 SDSS galaxies revealed a bimodal distribution in rest-frame $u-g$ colour, with GMM-fitted peaks at $u-g = 1.44$ (blue cloud) and $u-g = 1.92$ (red sequence), consistent with the colour separation reported by Strateva et al. (2001). ASTRA identified the bimodality autonomously, correctly interpreting it as reflecting two distinct stellar-population histories: ongoing star formation in blue-cloud galaxies and quenched, passively evolving stellar populations in the red sequence. The blue fraction declines from ~50 per cent at low stellar masses ($\log M_\star/M_\odot \lesssim 10$) to ~10 per cent at high masses ($\log M_\star/M_\odot \gtrsim 11$), reflecting the well-known mass-dependent quenching of star formation (Peng et al. 2010). This result validates ASTRA's unsupervised pattern recognition on complex, multi-dimensional datasets.

9.4 Confounder Detection

As part of the galaxy colour analysis, ASTRA's causal reasoning module performed automated confounder detection on the SDSS galaxy sample. The system identified u-band apparent magnitude as the strongest confounder influencing the colour–mass relationship, with a measured bias of 0.2351. This is physically expected: u-band flux is sensitive to both recent star formation (which drives blue colours) and distance (which affects apparent magnitude at fixed luminosity), making it a confounding variable in any analysis that does not explicitly control for survey depth. The automated detection of this confounder demonstrates ASTRA's ability to identify variables that could bias scientific conclusions if left uncontrolled, a capability that goes beyond simple correlation analysis.

9.5 HR Diagram and the Main Sequence

Using 4 984 stars from Gaia Early Data Release 3, ASTRA recovered the main-sequence relationship $M_G = 3.66 \times (G_{\rm BP} - G_{\rm RP})$ with a Pearson correlation $r = 0.90$ ($p < 10^{-100}$). The physics engine provided an interpretation via the Stefan–Boltzmann law: more luminous stars have higher effective temperatures, producing bluer colours and brighter absolute magnitudes. By recovering this textbook relationship from raw photometric data, ASTRA demonstrates that its statistical and physical reasoning modules operate correctly on large stellar samples.

9.6 Causal Direction Discovery

Applying the PC and FCI (Zhang 2008) causal discovery algorithms to a combined stellar–galaxy dataset, ASTRA correctly identified redshift as a cause of observed colour (cosmological reddening), rather than the reverse. This is a non-trivial result: in purely correlational analyses, redshift and colour are symmetric, but causal algorithms break this symmetry by exploiting conditional independence structure. ASTRA's correct recovery of the causal direction—cosmological effects driving observed photometric properties, not the reverse—validates the causal reasoning module on real multi-source data.

These five live-data results, obtained without manual tuning or domain-specific guidance, demonstrate that ASTRA generalises beyond its controlled test cases. The recovered relationships span classical mechanics (Kepler's law), cosmology (dark energy), galaxy evolution (colour bimodality), stellar astrophysics (HR diagram), and causal inference (redshift–colour direction), collectively exercising the full breadth of the framework's analytical capabilities.

10 CROSS-DOMAIN SCIENTIFIC VALIDATION

A critical test for any general-purpose analytical framework is whether its capabilities transfer beyond the domain for which it was originally developed. To evaluate this, we deployed ASTRA on publicly available datasets from three non-astrophysical domains—economics, climate science, and epidemiology—and performed a cross-domain synthesis linking variables across these fields. In each case, ASTRA operated without domain-specific guidance: the system ingested the data, autonomously identified patterns, applied its causal and statistical reasoning modules, and generated hypotheses that we compare with established results from the relevant literatures.

All data used in this section are drawn from public archives: the World Bank Open Data repository (World Bank 2024), NASA Goddard Institute for Space Studies (GISS) global temperature records (Hansen et al. 2010), and the NOAA Mauna Loa CO_2 time series (NOAA Global Monitoring Laboratory 2024).

10.1 Economics: World Bank Macroeconomic Indicators

ASTRA analysed macroeconomic indicators for 217 countries from the World Bank Open Data repository (World Bank 2024), spanning 1960–2023 and covering GDP, unemployment, trade balance, inflation, government debt, income inequality (Gini coefficient), and related variables. Operating in discovery mode, the system generated 10 hypotheses and tested each against the data.

Validated hypotheses (9/10):

(i) **Okun's Law:** ASTRA recovers a negative relationship between GDP growth and unemployment change, consistent with the empirical regularity reported by Okun (1962). The estimated coefficient (−1.8 per cent unemployment per 1 per cent GDP shortfall) is within the range reported in the literature.

(ii) **Trade–GDP correlation:** Trade openness (exports + imports as a fraction of GDP) is positively correlated with GDP per capita ($r = 0.41$, $p < 10^{-5}$), consistent with standard growth models.

(iii) **Gini inequality persistence:** Income inequality, as measured by the Gini coefficient, exhibits strong temporal persistence ($r = 0.92$ at 5-year lag), consistent with structural theories of inequality.

(iv) **PPP convergence:** Purchasing power parity shows convergence over decadal time-scales, consistent with the Balassa–Samuelson effect.

(v) **GDP mean-reversion:** GDP growth rates exhibit mean-reversion on 5–10 year time-scales, consistent with business cycle theory.

(vi) **Reinhart–Rogoff debt threshold:** ASTRA identifies a weakly negative relationship between government debt-to-GDP ratio and growth above the ~90 per cent threshold reported by Reinhart & Rogoff (2010), though the effect is not statistically robust ($p = 0.08$) and the system correctly flags the sensitivity to outlier countries and

the endogeneity concern (slow growth causes high debt, not only the reverse).

(vii) **Export diversification:** Economies with more diversified export baskets show lower GDP volatility ($r = -0.38$, $p < 0.001$).

(viii) **Inflation persistence:** Inflation rates exhibit strong autocorrelation ($r = 0.78$ at 1-year lag), consistent with adaptive expectations models.

(ix) **Finance–growth nexus:** Financial sector depth (domestic credit to private sector as a fraction of GDP) is positively associated with GDP per capita ($r = 0.52$, $p < 10^{-8}$), though the causal direction remains ambiguous.

Failed hypothesis (1/10):

(x) **Phillips Curve:** Astra tested for a negative relationship between unemployment and inflation, as originally reported by Phillips (1958). The system found no statistically significant relationship in the post-1990 cross-country data ($r = -0.03$, $p = 0.62$). This negative result is consistent with the well-documented breakdown of the simple Phillips Curve in modern macroeconomic data and demonstrates that Astra correctly identifies non-relationships rather than forcing spurious patterns.

The Phillips Curve failure is a particularly important result: it demonstrates that Astra does not simply confirm every hypothesis it generates. The system's ability to identify relationships that do *not* hold in the data is as scientifically valuable as its ability to identify those that do.

10.2 Climate Science: NASA GISS and NOAA Data

Astra analysed global mean surface temperature anomalies from NASA GISS (Hansen et al. 2010) and atmospheric CO_2 concentrations from the NOAA Mauna Loa Observatory (NOAA Global Monitoring Laboratory 2024), covering the period 1880–2023 (temperature) and 1958–2023 (CO_2).

Validated hypotheses (4/5):

(i) **CO_2–temperature correlation:** Astra identifies a strong positive correlation between atmospheric CO_2 concentration and global mean temperature anomaly ($R^2 = 0.936$ over the 1958–2023 overlap period). The system correctly notes that correlation does not establish causation but flags the temporal precedence of CO_2 increases as suggestive of a causal relationship.

(ii) **Warming acceleration:** The rate of warming post-1990 ($0.20\,^\circ\mathrm{C\,decade^{-1}}$) is 67 per cent faster than the pre-1990 rate ($0.12\,^\circ\mathrm{C\,decade^{-1}}$), consistent with accelerating radiative forcing from greenhouse gas accumulation.

(iii) **CO_2 growth acceleration:** The annual rate of CO_2 increase has itself accelerated: the mean annual increment was $\sim 1.0\,\mathrm{ppm\,yr^{-1}}$ in the 1960s, rising to $\sim 2.4\,\mathrm{ppm\,yr^{-1}}$ in the 2010s, consistent with increasing anthropogenic emissions.

(iv) **Decadal warming trend:** Each successive decade since the 1970s has been warmer than the preceding one, a pattern that Astra identifies as statistically significant at the $p < 0.001$ level using a non-parametric trend test.

Inconclusive hypothesis (1/5): The system tested for an El Niño–Southern Oscillation (ENSO) modulation signal in the temperature residuals after removing the long-term trend. While periodic structure was detected, Astra could not establish a statistically robust ENSO signal with the available temporal resolution, and correctly flagged the result as inconclusive.

10.3 Epidemiology: World Bank Health Indicators

Astra analysed health and development indicators for 217 countries from the World Bank (World Bank 2024), including life expectancy, infant mortality, GDP per capita, health expenditure, vaccination coverage, and maternal mortality.

Validated hypotheses (5/5):

(i) **Infant mortality vs. GDP:** Astra identifies a strong negative relationship between infant mortality rate and GDP per capita ($R^2 = 0.587$), with a logarithmic functional form providing the best fit—consistent with the diminishing marginal returns of wealth on health outcomes.

(ii) **Preston Curve:** Life expectancy increases with GDP per capita following a concave (logarithmic) relationship ($R^2 = 0.686$), recovering the classic Preston Curve (Preston 1975). The system correctly notes the flattening at high GDP, indicating saturation of the wealth–health relationship.

(iii) **Life expectancy vs. health spending:** Health expenditure per capita is positively associated with life expectancy ($r = 0.62$, $p < 10^{-12}$), though Astra's causal module flags the likely bidirectional relationship (wealthier countries both spend more on health and have longer life expectancies, with GDP as a common cause).

(iv) **DPT vaccination coverage:** DPT3 vaccination coverage is negatively correlated with infant mortality ($r = -0.65$, $p < 10^{-15}$), consistent with the established effectiveness of childhood vaccination programmes.

(v) **Maternal mortality:** Maternal mortality ratio is strongly negatively associated with GDP per capita ($r = -0.58$, $p < 10^{-10}$) and with skilled birth attendance ($r = -0.71$, $p < 10^{-18}$), consistent with the role of healthcare infrastructure in reducing maternal deaths.

10.4 Cross-Domain Synthesis

To test whether Astra can identify meaningful relationships *across* domains, we merged the economics, climate, and health datasets at the country–year level and applied the full analytical pipeline. The system generated 8 cross-domain hypotheses and applied Benjamini–Hochberg false discovery rate (FDR) correction (Benjamini & Hochberg 1995) at $q = 0.05$ to account for multiple testing.

Validated after FDR correction (5/8):

(i) **GDP–CO_2 coupling:** National GDP is positively correlated with per-capita CO_2 emissions ($r = 0.71$, $p_{\mathrm{adj}} < 10^{-10}$), reflecting the carbon intensity of economic activity.

(ii) **Life expectancy–CO_2:** Countries with higher CO_2 emissions per capita tend to have higher life expectancies ($r = 0.58$, $p_{\mathrm{adj}} < 10^{-6}$). Astra correctly identifies this as a confounded relationship (GDP drives both) rather than a causal link from emissions to health.

(iii) **Wealth–Health Nexus:** GDP per capita, health expenditure, and life expectancy form a strongly coupled triad ($R^2 = 0.700$ for the three-variable model), which Astra identifies as the strongest cross-domain relationship. The causal module suggests GDP $\rightarrow$ health expenditure $\rightarrow$ life expectancy as the primary causal pathway, with direct GDP $\rightarrow$ life expectancy effects mediated by nutrition, sanitation, and education.

(iv) **Urbanization–CO_2:** Urbanization rate is positively correlated with per-capita CO_2 emissions ($r = 0.49$, $p_{\mathrm{adj}} < 10^{-4}$), consistent with higher energy consumption in urban settings.

(v) **Renewables–GDP paradox:** Renewable energy share is *negatively* correlated with GDP per capita ($r = -0.42$, $p_{\mathrm{adj}} < 0.001$). Astra correctly interprets this counter-intuitive finding: poorer countries rely more heavily on traditional biomass (counted as re-

newable), while wealthier countries have historically derived more energy from fossil fuels. The system flags this as an example where naïve interpretation of a correlation would be misleading.

Lost after FDR correction (3/8): Three hypotheses (education–emissions coupling, democracy–health association, and trade openness–environmental quality) did not survive FDR correction at $q = 0.05$. These marginal relationships may be real but require larger samples or more careful confound control to establish.

The application of FDR correction is methodologically important: without multiple testing correction, the cross-domain analysis would have reported 8/8 validated hypotheses, a misleadingly optimistic result. Astra's automatic application of FDR correction demonstrates responsible statistical practice.

10.5 Negative Results and Epistemic Honesty

The cross-domain validation provides several examples of Astra correctly identifying non-relationships or qualifying ambiguous findings:

Phillips Curve failure: As discussed in Section 10.1, Astra found no statistically significant unemployment–inflation trade-off in post-1990 data ($r = -0.03$, $p = 0.62$). This is consistent with the well-documented instability of the Phillips Curve and demonstrates that Astra does not force patterns where none exist.

FDR attrition: Of 8 cross-domain hypotheses, 3 were lost after FDR correction. Astra reports these as "not validated" rather than attempting to salvage them through post-hoc analysis.

Confound identification: In 3 of the 5 validated cross-domain hypotheses, Astra's causal module identified confounding variables (primarily GDP as a common cause) and flagged the correlations as potentially non-causal. This is a critical capability: reporting a correlation between CO_2 emissions and life expectancy without noting the GDP confound would be scientifically irresponsible.

Reinhart–Rogoff qualification: The debt–growth relationship was validated only with substantial caveats ($p = 0.08$, sensitivity to outliers, endogeneity concerns), demonstrating nuanced rather than binary hypothesis assessment.

These negative and qualified results are as important as the positive validations. A system that confirms every hypothesis it generates would be useless; Astra's ability to distinguish signal from noise, and to qualify ambiguous findings, is essential for its role as a scientific tool.

10.6 Summary of Cross-Domain Results

Table 2 provides a comprehensive listing of all hypotheses tested across the five scientific domains.

*Reinhart–Rogoff threshold validated with substantial caveats: $p = 0.08$, sensitive to outliers, endogeneity concerns flagged.

In total, Astra tested approximately 42 hypotheses across five scientific domains, validating 36 (including 1 with caveats), failing 1 (Phillips Curve), finding 1 inconclusive (ENSO), and losing 3 after FDR correction. This 86 per cent validation rate—achieved without domain-specific tuning and across disciplines as diverse as astrophysics, macroeconomics, and epidemiology—provides strong evidence that Astra's analytical capabilities generalize beyond its original astrophysical domain.

11 DISCUSSION

11.1 What ASTRA Demonstrates

The six controlled astrophysical test cases, five live deployments, and cross-domain validation in this paper collectively analyse data drawn from more than a dozen independent sources spanning five scientific domains. The astrophysical analyses process more than 27 430 data points from nine sources (Herschel, Chandra, SDSS, Gaia, NASA Exoplanet Archive, Pantheon+, synthetic molecular clouds, MPA-JHU value-added catalogue, and Gaia EDR3), while the cross-domain validation draws on national-level panel data for 217 countries over six decades (World Bank), global temperature records spanning 143 years (NASA GISS), and atmospheric CO_2 measurements covering 65 years (NOAA Mauna Loa). Rather than repeating the individual results, we reflect here on what these collective demonstrations reveal about Astra's strengths and limitations as an integrated tool.

Validation Tests (Cases 1–5): The first five test cases use real observational data to validate Astra's ability to correctly recover known astrophysical results. These tests demonstrate that Astra can:

- Perform dimensional analysis and physical validation (Test Case 1)
- Fuse multi-wavelength data with proper uncertainty propagation (Test Case 2)
- Identify established astrophysical relationships through pattern recognition (Test Case 3)
- Distinguish causation from correlation using structural causal models (Test Case 4)
- Perform rigorous Bayesian model comparison with evidence computation (Test Case 5)

The integration advantage is most concretely demonstrated in Test Case 5 (Bayesian Model Selection), where physical motivation (the virial theorem predicting a power law) and statistical evidence (Bayes factors strongly favouring power-law and logarithmic models over linear alternatives) jointly inform model interpretation. Astra's framework correctly identifies that the power-law and logarithmic models are statistically indistinguishable by Bayesian evidence, then uses the physical motivation of the virial theorem to prefer the power-law form. This joint consideration of statistical evidence and physical constraints—within a single automated workflow rather than requiring manual synthesis of separate analyses—represents the core value of the integrated approach.

Discovery Test (Case 6): Test Case 6 demonstrates a different capability—discovery-mode operation beyond knowledge retrieval, demonstrated under controlled conditions. By operating in "knowledge isolation mode" on a synthetic dataset with embedded causal structure, Astra discovered the correct causal drivers (Jeans mass, magnetic field, virial parameter), identified column density as a proxy rather than direct cause, and generated testable predictions for future validation. This test addresses a key question: *can Astra go beyond "correct recovery of known results" to demonstrate genuine discovery capability?* The results suggest that Astra's architecture supports discovery-mode operation, though definitive scientific discovery requires collaboration with domain experts and observational validation.

Four Modes of Evidence:

(i) **Validation mode** (Tests 1–5): Astra analyses real data to recover known astrophysical relationships, validating that the integrated framework correctly implements established analysis methods.

(ii) **Discovery mode** (Test 6): Astra analyses data in knowledge

Table 2. Summary of all hypotheses tested across five scientific domains. "Validated" indicates the hypothesis is supported by the data at the stated significance level; "Failed" indicates the hypothesis is not supported; "FDR lost" indicates the hypothesis did not survive false discovery rate correction.

Domain	Hypothesis	Key Statistic	Valid?	Data Source
Astrophysics — Controlled Test Cases				
Astrophysics	Filament universal width	0.098 ± 0.019 pc	✓	Herschel
Astrophysics	Virial scaling relation	$r = 0.988$, $p < 10^{-18}$	✓	Herschel
Astrophysics	Multi-λ cross-match	60/370 secure matches	✓	CDFS
Astrophysics	Mass–metallicity relation	$r = 0.68$, $p < 10^{-27}$	✓	SDSS
Astrophysics	Environmental quenching	ΔSFR ≈ -0.3 dex	✓	SDSS
Astrophysics	Causal structure (Gaia)	Correct DAG recovery	✓	Gaia DR2
Astrophysics	Bayesian model selection	BF = 1.2 (PL vs. log)	✓	Herschel
Astrophysics	SF causal drivers (synthetic)	3/3 drivers found	✓	Synthetic
Astrophysics	Column density as proxy	Proxy confirmed	✓	Synthetic
Astrophysics — Live Deployments				
Astrophysics	Kepler's third law	$R^2 = 0.9982$	✓	NASA Exo. Archive
Astrophysics	Dark energy signature	$\chi^2/\mathrm{dof} = 0.64$	✓	Pantheon+
Astrophysics	Galaxy colour bimodality	Peaks: 1.44, 1.92	✓	SDSS
Astrophysics	u-band confounder	Bias = 0.2351	✓	SDSS
Astrophysics	HR main sequence	$r = 0.90$, $p < 10^{-100}$	✓	Gaia EDR3
Astrophysics	Causal direction ($z \to$ colour)	Correct orientation	✓	Gaia+SDSS
Economics				
Economics	Okun's Law	$-1.8\%/1\%$ GDP	✓	World Bank
Economics	Trade–GDP correlation	$r = 0.41$, $p < 10^{-5}$	✓	World Bank
Economics	Gini persistence	$r = 0.92$ (5-yr lag)	✓	World Bank
Economics	PPP convergence	Decadal convergence	✓	World Bank
Economics	GDP mean-reversion	5–10 yr cycle	✓	World Bank
Economics	Reinhart–Rogoff threshold	$p = 0.08$ (qualified)	✓*	World Bank
Economics	Export diversification	$r = -0.38$, $p < 0.001$	✓	World Bank
Economics	Inflation persistence	$r = 0.78$ (1-yr lag)	✓	World Bank
Economics	Finance–growth nexus	$r = 0.52$, $p < 10^{-8}$	✓	World Bank
Economics	Phillips Curve	$r = -0.03$, $p = 0.62$	×	World Bank
Climate Science				
Climate	CO_2–temp. correlation	$R^2 = 0.936$	✓	GISS + NOAA
Climate	Warming acceleration	67% faster post-1990	✓	GISS
Climate	CO_2 growth acceleration	$1.0 \to 2.4$ ppm yr^{-1}	✓	NOAA
Climate	Decadal warming trend	$p < 0.001$	✓	GISS
Climate	ENSO modulation	Inconclusive	–	GISS
Epidemiology				
Epidemiology	Infant mortality vs. GDP	$R^2 = 0.587$	✓	World Bank
Epidemiology	Preston Curve	$R^2 = 0.686$	✓	World Bank
Epidemiology	Life exp. vs. health spending	$r = 0.62$, $p < 10^{-12}$	✓	World Bank
Epidemiology	DPT vaccination effect	$r = -0.65$, $p < 10^{-15}$	✓	World Bank
Epidemiology	Maternal mortality	$r = -0.71$, $p < 10^{-18}$	✓	World Bank
Cross-Domain Synthesis (after FDR correction)				
Cross-domain	GDP–CO_2 coupling	$r = 0.71$, $p_{\mathrm{adj}} < 10^{-10}$	✓	WB + NOAA
Cross-domain	Life exp.–CO_2 (confounded)	$r = 0.58$, $p_{\mathrm{adj}} < 10^{-6}$	✓	WB + NOAA
Cross-domain	Wealth–Health Nexus	$R^2 = 0.700$	✓	World Bank
Cross-domain	Urbanization–CO_2	$r = 0.49$, $p_{\mathrm{adj}} < 10^{-4}$	✓	WB + NOAA
Cross-domain	Renewables–GDP paradox	$r = -0.42$, $p_{\mathrm{adj}} < 0.001$	✓	World Bank
Cross-domain	Education–emissions	Not significant after FDR	FDR lost	WB + NOAA
Cross-domain	Democracy–health	Not significant after FDR	FDR lost	World Bank
Cross-domain	Trade–environment	Not significant after FDR	FDR lost	World Bank

isolation mode to discover patterns without being told what to look for, generating hypotheses and testable predictions.

(iii) **Live deployment** (Section 9): Astra operates on large archival datasets without manual guidance, independently recovering fundamental results spanning classical mechanics, cosmology, galaxy evolution, stellar astrophysics, and causal inference.

(iv) **Cross-domain generalization** (Section 10): Astra operates on non-astrophysical data from economics, climate science, and epidemiology, recovering established relationships and correctly identifying non-relationships—demonstrating that the framework's analytical capabilities are not domain-specific artefacts.

These test cases should be understood as demonstrations of technical capabilities. For Tests 1–5, the value lies in showing Astra can correctly recover known results using an integrated framework. For Test 6, the value lies in showing Astra can work in discovery mode,

though the predictions generated require observational validation. For the cross-domain validation, the value lies in demonstrating that Astra's analytical architecture generalizes beyond astrophysics.

11.2 What ASTRA Genuinely Contributes

A fair question is whether Astra merely confirms what is already known. The answer is nuanced.

The live validation results (Section 9), while impressive for a fully automated system, recover well-established physics—Kepler's third law, the Hubble diagram, galaxy colour bimodality, the HR main sequence. These results should not be over-interpreted as novel discoveries; their value lies in demonstrating that Astra's integrated pipeline produces correct, quantitatively precise results on real data without human intervention. That an automated system can ingest raw archival data and, without being told what to seek, arrive at $R^2 = 0.9982$ for Kepler's law or $\chi^2/\mathrm{dof} = 0.64$ for the Hubble diagram is a non-trivial engineering and methodological achievement, even though the underlying physics is textbook material.

The cross-domain results (Section 10) extend this argument: recovering Okun's Law, the Preston Curve, and CO_2–temperature coupling from raw public data—using the same analytical pipeline that recovers Kepler's law and galaxy bimodality—demonstrates that Astra's capabilities are genuinely domain-general rather than astrophysics-specific. The Phillips Curve failure and FDR attrition of cross-domain hypotheses provide further evidence that the system produces honest, calibrated results rather than indiscriminate confirmations.

Astra's genuine scientific contribution lies in four areas. First, the *integration itself*: by chaining dimensional analysis, causal inference, and Bayesian model selection into a single workflow, Astra eliminates inconsistencies that arise when scientists manually combine outputs from separate tools—particularly in uncertainty propagation and physical validation. Second, the *causal reasoning capability*: Test Case 6 demonstrates that Astra can distinguish proxy variables from causal drivers, a distinction that purely correlational or ML-based approaches cannot make. Third, the *knowledge isolation mode*: by disabling access to encoded domain knowledge, Astra provides a controlled test of whether patterns in data are discoverable from first principles, an approach that could be applied to domains where the ground truth is genuinely unknown. Fourth, the *cross-domain generalization*: the same analytical pipeline that recovers astrophysical scaling relations also recovers macroeconomic regularities, climate trends, and epidemiological relationships, with appropriate epistemic qualifications including confound identification, FDR correction, and negative result reporting.

11.3 Cross-Domain Generalization

The cross-domain validation (Section 10) addresses a fundamental question about Astra's design: is the framework genuinely general-purpose, or are its capabilities specific to the astrophysical domain for which it was developed?

Three aspects of the cross-domain results support generality. First, the *same analytical modules*—dimensional analysis, causal inference, Bayesian model selection, pattern recognition—that recover astrophysical relationships also recover economic, climatic, and epidemiological relationships, without any domain-specific modifications. Second, Astra's *epistemic qualifications* transfer across domains: the system correctly identifies GDP as a confound in the life expectancy–CO_2 correlation, flags the Reinhart–Rogoff threshold as sensitive to outliers and endogeneity, and applies FDR correction to the cross-domain synthesis—demonstrating that its causal reasoning and statistical rigour are not limited to astrophysical contexts. Third, the *negative results*—the Phillips Curve failure, the ENSO inconclusive result, and the three FDR-lost cross-domain hypotheses—demonstrate that Astra does not indiscriminately confirm hypotheses, even in domains where the system has no encoded knowledge to guide it toward "correct" answers.

These results suggest that Astra's architecture is genuinely domain-agnostic at the analytical level, even though the system was originally developed for astrophysics. The core capabilities—pattern discovery, causal inference, hypothesis testing with multiple testing correction, confound identification—are domain-general by construction, since they rely on statistical and causal methods that are not specific to any scientific field.

Caveats: The cross-domain validation uses well-established relationships as benchmarks. Astra has not been tested on genuinely open questions in economics, climate science, or epidemiology, and its performance on such problems remains unknown. The domain knowledge systems (MORK ontology, 75 specialized modules) are astrophysics-specific; extending Astra to production-quality analysis in other domains would require developing analogous domain knowledge for those fields. The cross-domain results should be understood as evidence that the *analytical architecture* generalizes, not that Astra is ready for deployment as an economics or climate science tool.

11.4 Positioning Relative to Other Approaches

Astra occupies a distinct position in the computational astronomy ecosystem by integrating multiple analysis types within a unified framework. Rather than replacing existing tools, Astra aims to provide consistent interfaces and workflows for complex multi-step analyses that would otherwise require manual combination of separate pipelines.

Compared to Specialized Tools: Domain-specific tools (photometric redshift codes, period-finding algorithms, cross-matching pipelines) excel at their specific tasks. Astra does not aim to replace these tools but to provide integrated workflows that combine multiple analysis types with consistent uncertainty handling and physical validation.

Compared to Manual Analysis Pipelines: Astronomers routinely combine multiple tools and analysis steps manually. This manual integration creates opportunities for inconsistent uncertainty treatment, missed physical constraints, and reproducibility challenges. Astra addresses these by providing automated workflows with integrated validation.

Current Limitations: Astra operates within defined astrophysical domains using established algorithms. The system assists scientific reasoning rather than replacing domain expertise. Results require interpretation and validation by domain scientists. The test cases presented use small samples and should be understood as capability demonstrations rather than production-scale analyses.

11.5 ASTRA's Role in Scientific Discovery

A crucial question is: *what role can Astra play in genuine scientific discovery?* We would not expect to give Astra one of the big contemporary problems like "resolve the nature of the Hubble Tension," or "tell us what dark matter is," or "prove that we live in a holographic universe," and expect to get definitive answers in a computational Eureka moment. These problems require deep physical insight, creative

hypothesis formation, and careful experimental design—capabilities that remain fundamentally human.

However, Astra's discovery architecture, used alongside and by an experienced astronomer, has capabilities in discovery and inference that go beyond those of straightforward AI (large language models) and/or machine learning. Test Case 6 demonstrates that Astra can:

- Discover patterns in data without being told what to look for (knowledge isolation mode)
- Apply causal inference methods that distinguish correlation from causation
- Generate competing hypotheses and rank them by multiple criteria (hypothesis scores reflect fit to correlational data and should not be interpreted as definitive causal selection)
- Produce testable predictions that can guide future observational programmes

Astra's role in science is to work alongside the experienced astronomer to analyse data and facilitate genuine discovery. Astra is therefore a tool to assist the astronomer, rather than a replacement for domain expertise. The human scientist brings deep physical understanding, creative insight, and judgment about which questions are worth asking. Astra brings automated pattern discovery, causal reasoning, and systematic hypothesis evaluation. Together, they can pursue scientific questions more effectively than either could alone.

This collaborative model—Astra as discovery assistant working alongside domain experts—is the appropriate way to frame AI-assisted scientific discovery. Claims that AI systems will autonomously resolve major open questions in physics overlook the essential role of human physical insight and the importance of careful experimental design. Astra's architecture supports discovery-mode operation, but definitive scientific discovery requires collaboration with domain experts and observational validation.

11.6 Limitations and Future Work

Current limitations include:

- **Sample sizes:** The controlled test cases use 24–1000 objects, which are small for production analyses. The live deployments reach larger scales (up to 4 984 stars), but scaling to survey-scale datasets (10^6–10^9 objects) requires further validation of computational performance and statistical robustness.
- **Computational cost:** Causal discovery scales as $O(p^2)$ for p variables, limiting applicability to high-dimensional problems.
- **Prior dependence:** Bayesian evidence computation requires careful prior specification, and results can be prior-dependent.
- **Domain knowledge:** Astra's pattern recognition retrieves encoded domain knowledge. Genuine novel discovery requires identifying patterns not already in the knowledge base.
- **Selection effects:** Multi-wavelength cross-matching assumptions (Gaussian errors, uniform priors) may not hold in all regimes.
- **Live validation scope:** The live deployment results, while quantitatively precise, recover well-established physics and should not be over-interpreted as evidence of discovery capability. They validate the pipeline's correctness, not its ability to find genuinely new science.
- **Cross-domain depth:** The cross-domain validation uses well-established benchmarks and has not been tested on open questions in non-astrophysical domains. Production-quality analysis in economics, climate science, or epidemiology would require domain-specific knowledge modules analogous to Astra's astrophysical MORK ontology.

Future work should focus on: scaling to larger datasets; validation on more scientifically challenging problems where the answer is not known in advance; integration with time-domain surveys for real-time analysis; extension to gravitational wave and neutrino multi-messenger astronomy; applying knowledge-isolation discovery mode to real datasets where causal structure is not known a priori; and developing domain knowledge modules for non-astrophysical fields to enable production-quality cross-domain analysis.

12 CONCLUSION

We have presented Astra (Autonomous System for Scientific Discovery in Astrophysics), an integrated framework for physics-aware scientific analysis. Through six controlled astrophysical test cases, five live deployments on archival data, and cross-domain validation across economics, climate science, and epidemiology, we have demonstrated Astra's technical capabilities across approximately 42 hypotheses in five scientific domains.

Test 1: Scaling Relations Analysis (24 Herschel filaments)

- Identified universal width: 0.098 ± 0.019 pc
- Measured virial scaling with $r = 0.988$, $p < 10^{-18}$
- Found 2.7σ tension with theoretical prediction, warranting further investigation

Test 2: Multi-Wavelength Data Fusion (CDFS, 60/370 sources)

- Bayesian cross-matching with probabilistic associations
- Classified 41 AGN, 19 stars, 0 galaxies (selection effect discussed)
- Proper astrometric uncertainty propagation

Test 3: Pattern Recognition (600 SDSS galaxies)

- Identified established astrophysical relationships: mass–metallicity relation, environmental quenching
- Demonstrated pattern recognition capability retrieving known literature results
- Not novel discovery but validation of pattern recognition implementation

Test 4: Causal Inference (1000 Gaia stars)

- Correctly recovered textbook causal relationships (validation demonstrator)
- Distinguished physical laws from selection biases
- Acknowledged this is a basic demonstration, not novel discovery

Test 5: Bayesian Model Selection (24 filaments)

- Power law and logarithmic models statistically indistinguishable (Bayes factor 1.2)
- Both strongly preferred over linear and broken power-law models
- Detected 2.7σ tension with virial theorem prediction
- Weakly-informative priors specified for reproducibility

Test 6: Discovery-Mode Operation (synthetic molecular clouds, $N = 500$)

- Discovered three causal drivers (Jeans mass, magnetic field, virial parameter) in knowledge isolation mode
- Correctly identified column density as proxy rather than direct cause

- Generated four testable predictions for future observational validation
- Performed intervention analysis validating causal claims against ground truth
- Demonstrated discovery-mode capability under controlled conditions

Live Deployment (Section 9)

- Recovered Kepler's third law from 2 839 exoplanets (R^2 = 0.9982, multivariate fit capturing stellar-mass dependence)
- Recovered dark energy signature from 1 701 SNe Ia (χ^2/dof = 0.64)
- Detected galaxy colour bimodality from 2 000 SDSS galaxies ($u-g$ peaks at 1.44 and 1.92; blue fraction declining with stellar mass)
- Recovered HR diagram main sequence from 4 984 Gaia stars (r = 0.90)
- Correctly inferred causal direction of redshift → colour
- Detected u-band magnitude as strongest confounder (bias = 0.2351)

Cross-Domain Validation (Section 10)

- Economics: 9/10 hypotheses validated (Okun's Law, trade–GDP, inequality persistence, and 6 others); Phillips Curve correctly identified as non-significant
- Climate science: 4/5 hypotheses validated (CO_2–temperature R^2 = 0.936, warming acceleration, CO_2 growth, decadal trend)
- Epidemiology: 5/5 hypotheses validated (Preston Curve R^2 = 0.686, infant mortality, vaccination, maternal mortality, health spending)
- Cross-domain synthesis: 5/8 hypotheses survived FDR correction (Wealth–Health Nexus R^2 = 0.700 the strongest); 3 lost after multiple testing correction

Four Modes of Evidence: Tests 1–5 validate Astra's ability to correctly recover known astrophysical results using real data (*validation mode*). Test 6 demonstrates Astra's discovery-mode operation under controlled conditions, including knowledge isolation pattern discovery and intervention analysis to distinguish correlation from causation (*discovery mode*). The five live deployments confirm that the framework generalises to archival data at scale, independently recovering fundamental physics spanning classical mechanics, cosmology, galaxy evolution, and stellar astrophysics (*live deployment mode*). The cross-domain validation demonstrates that Astra's analytical capabilities—pattern discovery, causal inference, hypothesis testing, confound identification—generalize beyond astrophysics to economics, climate science, and epidemiology (*cross-domain generalization mode*).

Astra provides an integrated framework where physical validation, uncertainty quantification, causal reasoning, and Bayesian model selection are applied consistently throughout multi-step analyses. While individual components use established algorithms, their integration within a single framework enables analyses that would otherwise require manual combination of multiple separate tools and pipelines. The cross-domain results suggest that this integration is genuinely domain-general, opening the possibility of applying Astra's analytical architecture to scientific problems beyond astrophysics.

The complete source code, documentation, and reproducible analysis notebooks are available in our public GitHub repository (White & Dey 2026). We encourage community feedback, contribution, and validation of these results on larger and more diverse datasets.

ACKNOWLEDGEMENTS

We thank the anonymous referee for constructive comments that have significantly improved this manuscript. This work uses data from Gaia DR2 and EDR3, Herschel Gould Belt Survey, HST ACS, Chandra Deep Field South, SDSS, the NASA Exoplanet Archive, and the Pantheon+ supernova compilation. The cross-domain validation uses data from the World Bank Open Data repository (https://data.worldbank.org), the NASA Goddard Institute for Space Studies (GISS) Surface Temperature Analysis, and the NOAA Global Monitoring Laboratory Mauna Loa CO_2 record. We acknowledge the use of community software tools including causal-learn, dowhy, NumPy, SciPy, and scikit-learn.

DATA AVAILABILITY

The complete Astra source code and analysis notebooks are available at https://github.com/Tilanthi/ASTRA. All data used in this paper are publicly available from the original survey archives as cited in the text. World Bank data are available at https://data.worldbank.org; NASA GISS temperature data at https://data.giss.nasa.gov/gistemp/; NOAA Mauna Loa CO_2 data at https://gml.noaa.gov/ccgg/trends/.

REFERENCES

Abazajian K. N., Adelman-McCarthy J. K., Agüeros M. A., et al., 2009, ApJS, 182, 543
Alexander D. M., Bauer F. E., Brandt W. N., et al., 2003, AJ, 126, 539
André P., Men'shchikov A., Bontemps S., Hennemann M., Motte F., Schneider N., 2010, A&A, 518, L102
Arzoumanian D., André P., Didelon P., et al., 2011, A&A, 529, L6
Arzoumanian D., André P., Men'shchikov A., Könyves V., Schneider N., Motte F., 2019, A&A, 621, A114
Bauer F. E., Alexander D. M., Brandt W. N., Hornschemeier A. E., Garmire G. P., Schneider D. P., 2004, AJ, 128, 2048
Benjamini Y., Hochberg Y., 1995, Journal of the Royal Statistical Society: Series B, 57, 289
Brinchmann J., Charlot S., White S. D. M., Tremonti C., Kauffmann G., Heckman T., Kennicutt R., 2004, MNRAS, 351, 1151
Brout D., et al., 2022, ApJ, 938, 110
Buckingham E., 1914, Physical Review, 4, 345
Dressler A., 1980, ApJ, 236, 351
Gaia Collaboration Brown A. G. A., Vallenari A., Prusti T., de Bruijne J. H. J., Babusiaux C., Bailer-Jones C. A. L., 2018, A&A, 616, A1
Giacconi R., Zirm A., Wang J., Rosati P., Gilli R., Nonino M., Czoske O., 2001, ApJ, 551, 624
Hansen J., Ruedy R., Sato M., Lo K., 2010, Reviews of Geophysics, 48, RG4004
Kass R. E., Raftery A. E., 1995, Journal of the American Statistical Association, 90, 773
Meng X.-L., Wong W. H., 1996, Statistica Sinica, 6, 831
NOAA Global Monitoring Laboratory 2024, Trends in Atmospheric Carbon Dioxide: Mauna Loa CO_2 annual mean data, https://gml.noaa.gov/ccgg/trends/
Okun A. M., 1962, Proceedings of the Business and Economic Statistics Section, American Statistical Association, pp 98–104
Panopoulou G. V., Psarouli I., Taqi N., Andrews J. J., 2022, A&A, 657, L13
Pearl J., 2009, Causality: Models, Reasoning, and Inference, 2nd edn. Cambridge University Press, Cambridge
Peng Y.-j., Lilly S. J., Kovač K., Bosz E., Carollo C. M., McGee S. L., Alonso M. S., Tassis K., 2010, ApJ, 721, 193
Perlmutter S., Aldering G., Goldhaber G., et al., 1999, ApJ, 517, 565
Phillips A. W., 1958, Economica, 25, 283

Preston S. H., 1975, Population Studies, 29, 231
Reinhart C. M., Rogoff K. S., 2010, American Economic Review, 100, 573
Riess A. G., Filippenko A. V., Challis P., et al., 1998, AJ, 116, 1009
Schwarz G., 1978, Annals of Statistics, 6, 461
Spirtes P., Glymour C., Scheines R., 2000, Causation, Prediction, and Search, 2nd edn. MIT Press, Cambridge, MA
Strateva I., Ivezić Ž., Knapp G. R., Heckman T. M., Strauss M. A., Gunn J. E., Lupton R. H., 2001, AJ, 122, 1861
Sutherland W., Saunders W., 1992, MNRAS, 259, 413
Tremonti C. A., Heckman T. M., Kauffmann G., et al., 2004, ApJ, 613, 898
White G. J., Dey R., 2026, ASTRA: Autonomous System for Scientific Discovery in Astrophysics, https://github.com/Tilanthi/ASTRA
World Bank 2024, World Bank Open Data, https://data.worldbank.org
Zhang J., 2008, Artificial Intelligence, 172, 1873